UNIVERSITY of MARYLAND
MARLENE AND STEWART GREENEBAUM
COMPREHENSIVE CANCER CENTER

KAMUZU
UNIVERSITY
OF HEALTH SCIENCES

"Why not both?" A semi-supervised pipeline for a comprehensive and scalable analysis of immune heterogeneity in human samples

David Rach, Nginache Nampota-Nkomba, Godfrey Mvula, Felix A. Mkandawire, Osward M. Nyirenda, Winter A. Okoth, Daniela Franco, Andrea G. Buchwald, Franklin R. Toapanta, Marcelo B. Sztein, Miriam K. Laufer, **Kirsten E. Lyke**, **Cristiana Cairo**.

University of Maryland School of Medicine

October 28, 2026

Slides

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Spectral Flow Cytometry is amazing!!!

- Capable of resolving remarkably similar fluorophores → **increased number of markers per panel** compared to conventional flow cytometry (CFC)
- Allows for rapid acquisition of millions of events in a few minutes → typically **greater number of acquired cells per sample** compared to mass cytometry (MC)
- Spectral Flow Cytometry (SFC) datasets have an unique combination of **width and depth** compared to preceding technologies



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Spectral Flow Cytometry is amazing!!!



- Able to efficiently acquire greater number of cells → more thorough profiling of **rare cell subsets**
- Build one panel, evaluate multiple cell types → invaluable for studies with **limited biospecimen**
- Expand our knowledge about immune system heterogeneity

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Trade-off: Increasing analytical complexity

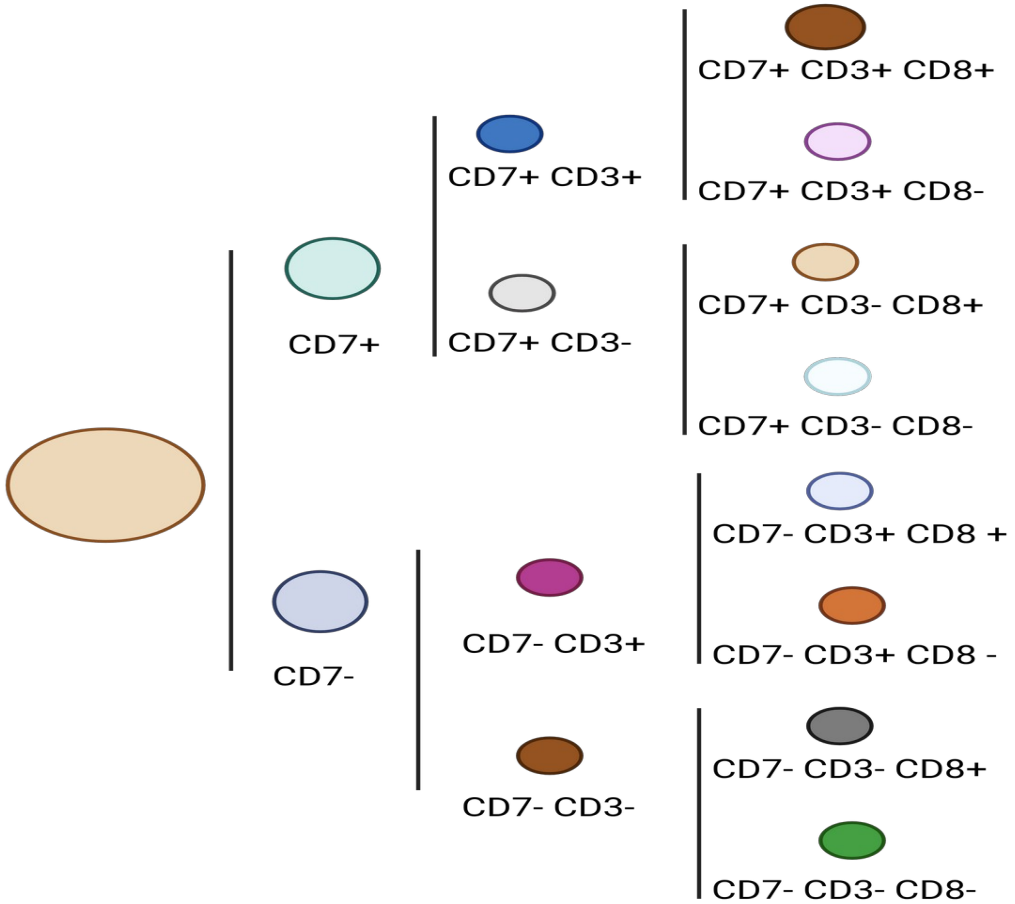
- If you are only interested in a **single population** of cells → analysis process remains <u>essentially unchanged</u> (manual gating with a few more gates)

- If your goal is to characterize **all cell populations** that may be present → the breadth and depth of the datasets complicate the analysis



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Complex Datasets: Combinatorially Exponential Clusters

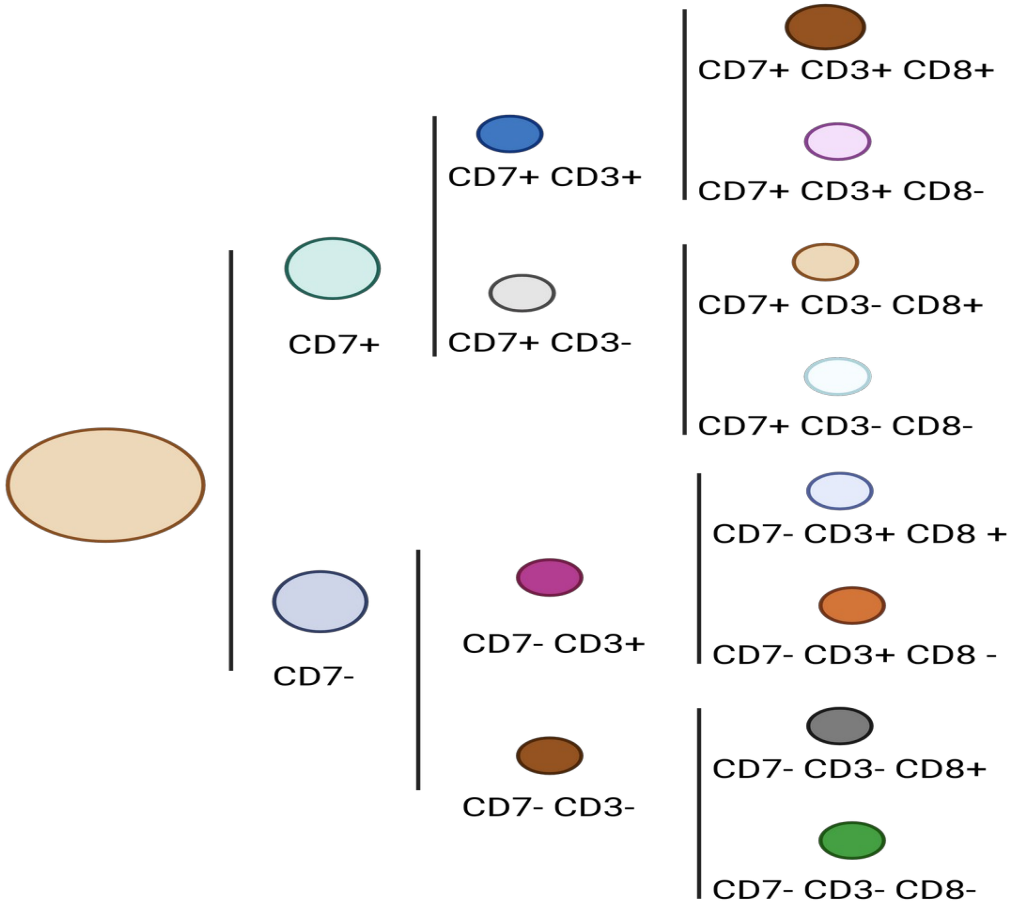


- If for a given marker in your panel, you **split** the cells into **positive** and **negative** populations
- And this splitting process was **repeated for every marker** in your panel
- At the end, we will have **grouped cells** that only **expressed the same markers**

- "Groups" per panel = $2^{\text{Number of Fluorophores}}$

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Complex Datasets: Combinatorially Exponential Clusters



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- **9-color** CFC panel → **512** groups at most
- **40-color** SFC panel → **1.1e+12** groups (**trillion**)
- **60-color** SFC panel → **1.2e+18** groups (**quintillion**)

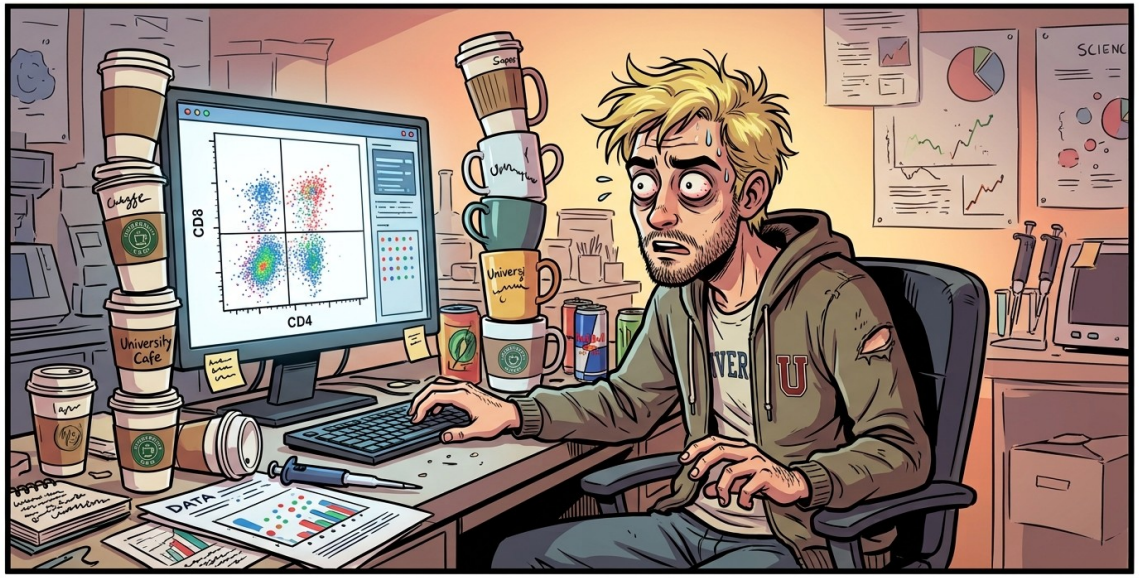
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What even is a quintillion?

If I were to manually check for every combination...

- **60-color** SFC panel → **1.2e+18** groups (**quintillion**)

  - 1 gate, every 10 seconds → 6 groups/minute
  - 60 minutes in an hour → 360 groups/hour
  - Awake 16 hours a day → 5760 groups/day
  - 365 days per year → **2,102,400 groups/year**



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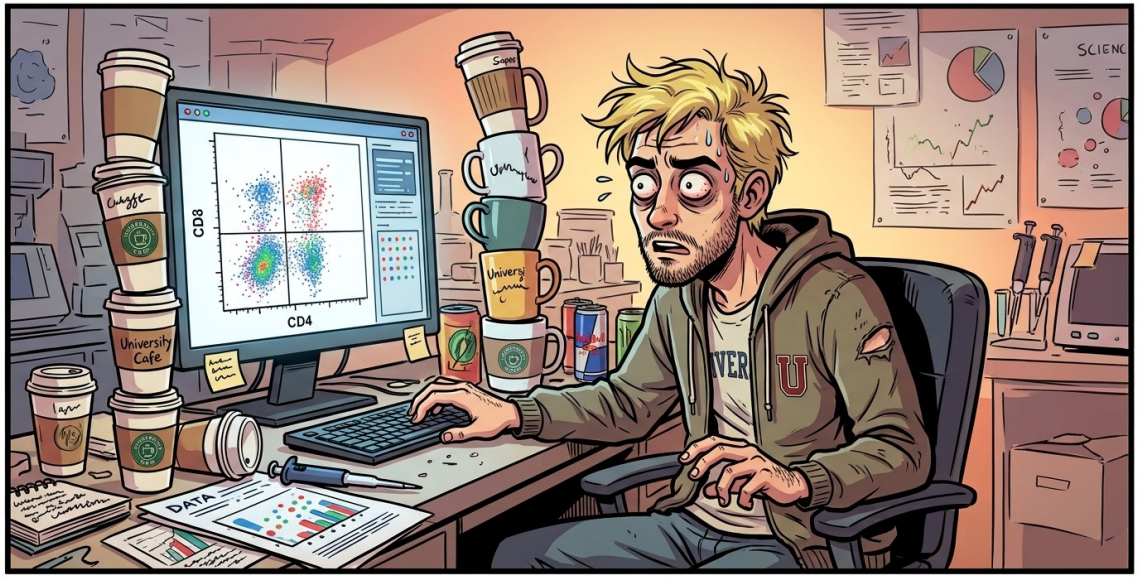
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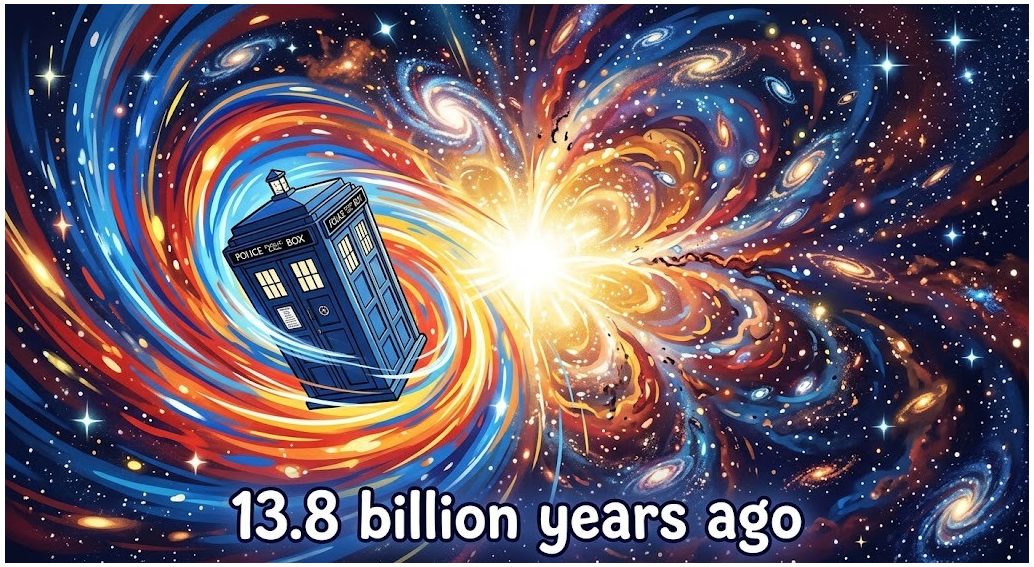
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  - Awake 16 hours a day → 5760 groups/day
  - 365 days per year → **2,102,400 groups/year**

….Not even close….



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Continuously since the Big Bang



- **60-color** SFC panel → **1.2e+18** groups (**quintillion**)

  - 365 days per year → **2,102,400 groups/year**
  - Invent a time-machine, become immortal
  - 13.8 billion years → **2.9e+16 groups/reality**

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Continuously since the Big Bang



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  - 13.8 billion years → **2.9e+16 groups/reality**

  - So we need ~ <u>50 immortal researchers working continuously since the Big Bang</u> just to check all group combinations for a **<u>single 60-fluorophore panel .fcs file</u>**

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Continuously since the Big Bang



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  - So we need ~ <u>50 immortal researchers working continuously since the Big Bang</u> just to check all group combinations for a **<u>single 60-fluorophore panel .fcs file</u>**

1 file down, 99 more to go!

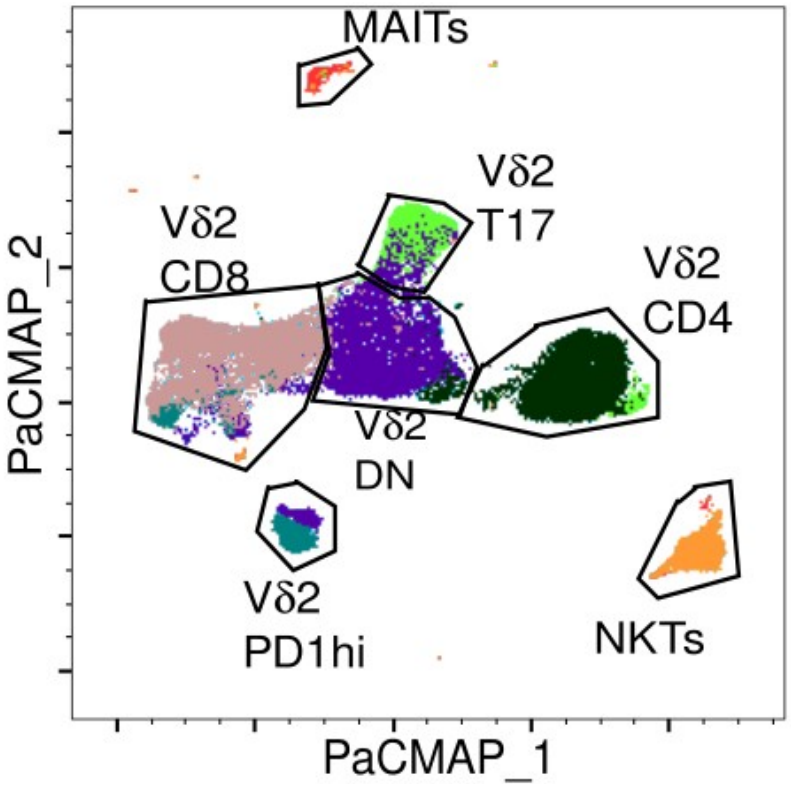
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Alternative: Unsupervised Analysis (Algorithms)

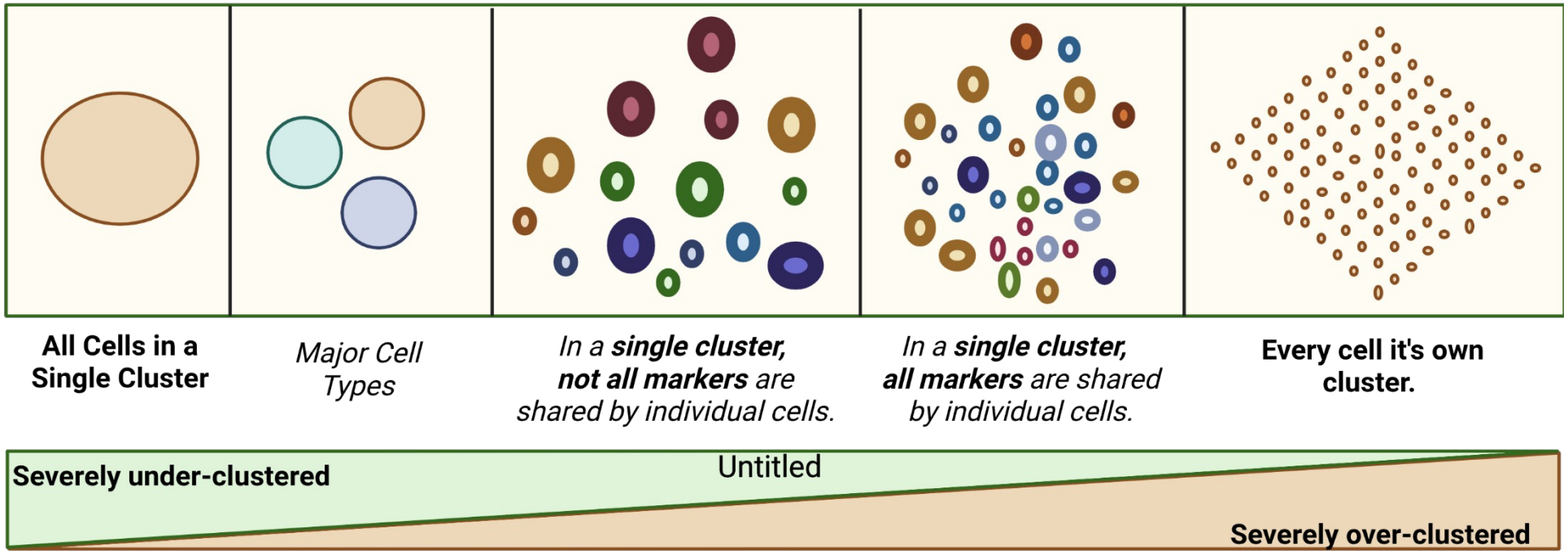
- Consequently, if the goal is to characterize all cell populations present, some form of **unsupervised (algorithm) or semi-supervised** analytical approaches will be needed

- **Clustering** → FlowSOM, Phenograph, Xshift, etc.
- **Dimensionality visualization** → opt-SNE, UMAP, PaCMAP, Phate, etc.

- **Automated Gating** → openCyto, flowMagic, ElasticGate, etc.

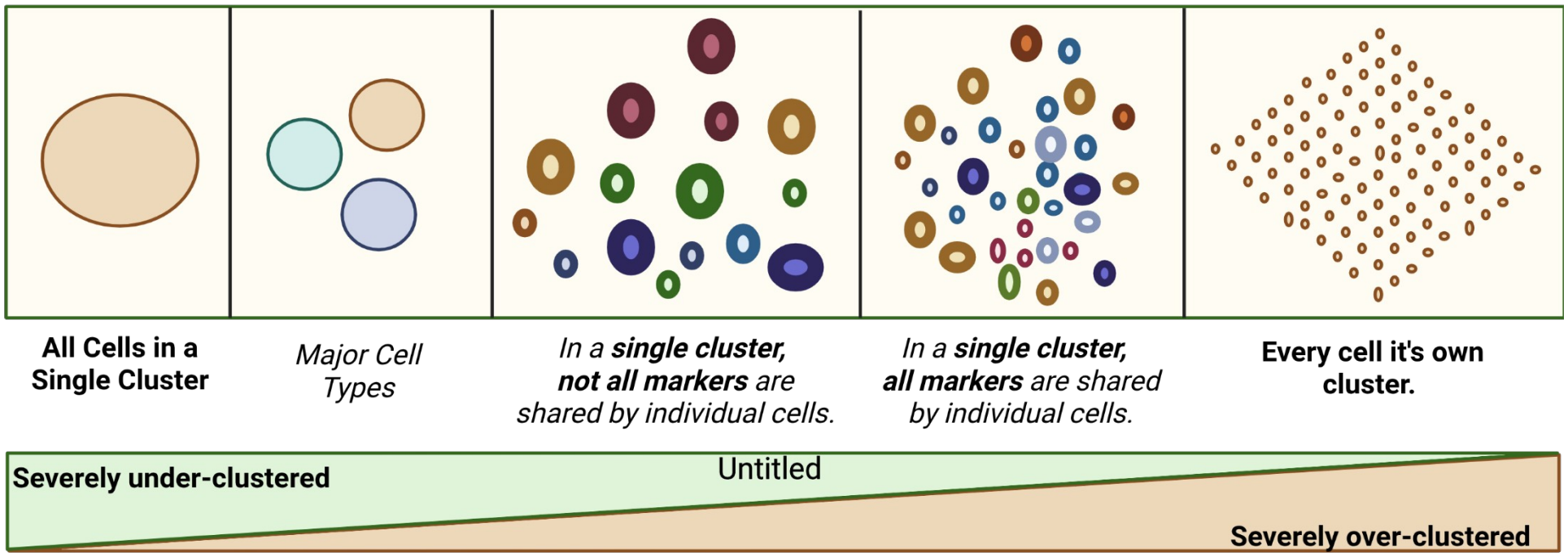


The challenge for unsupervised approaches



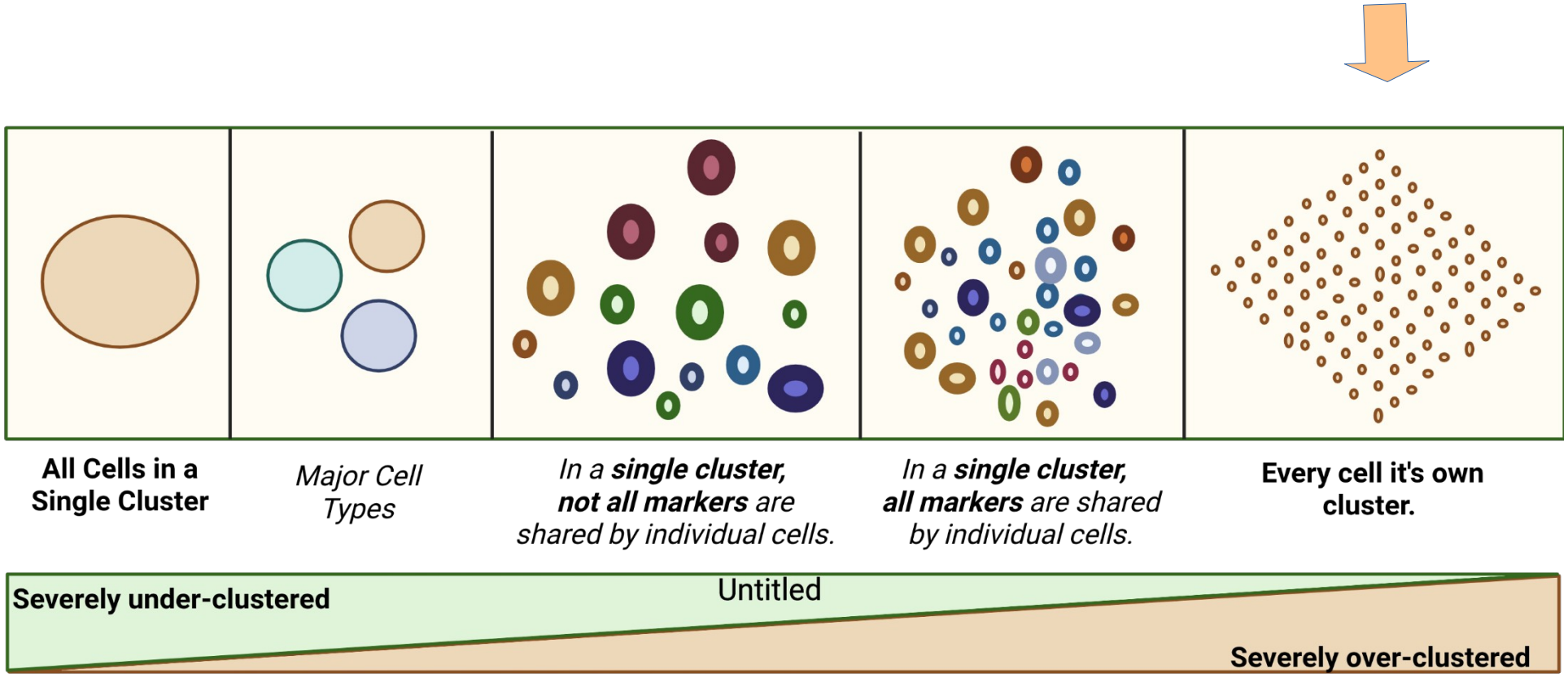
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The challenge for unsupervised approaches



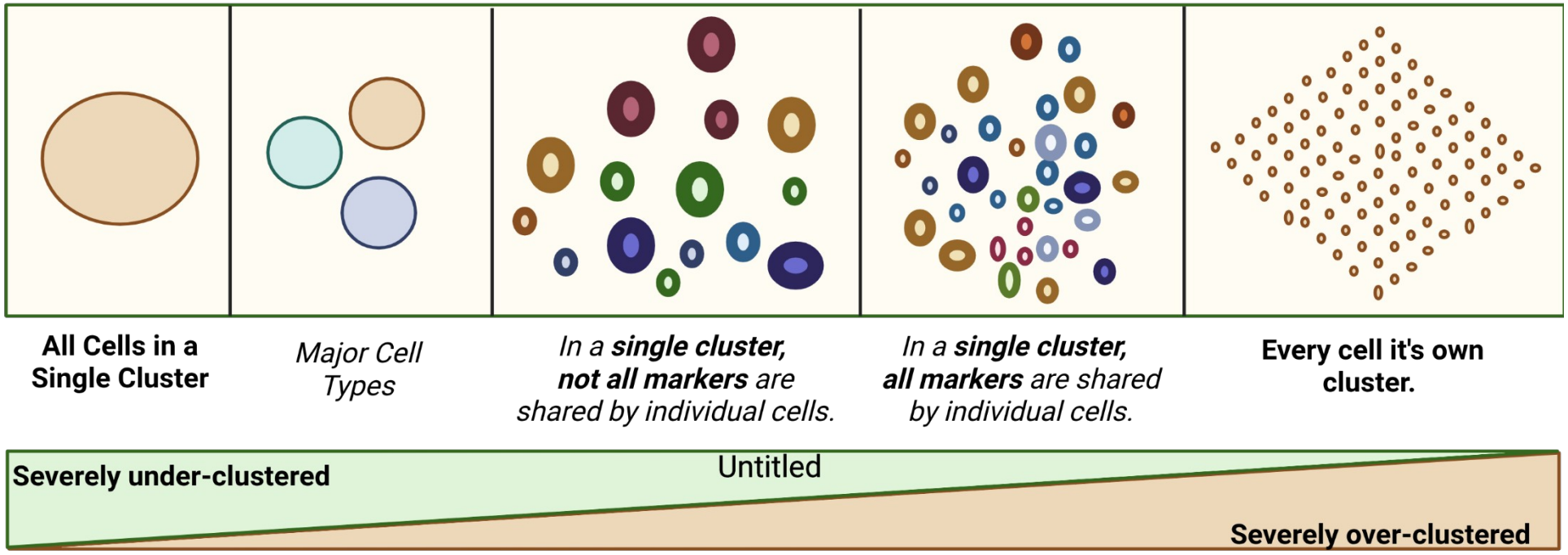
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The challenge for unsupervised approaches



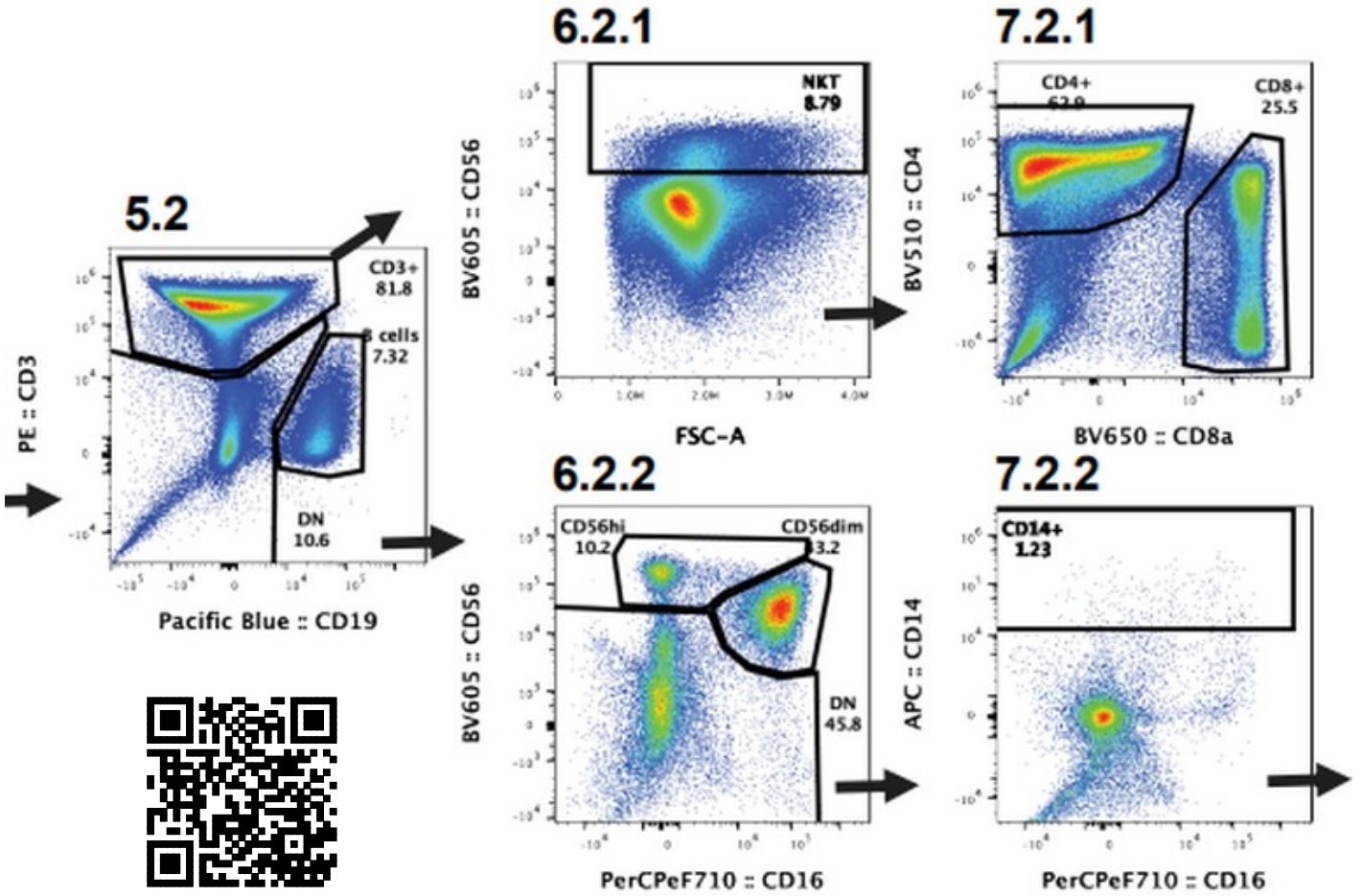
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The challenge for unsupervised approaches



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Many algorithmic tools struggle with SFC data's quirks



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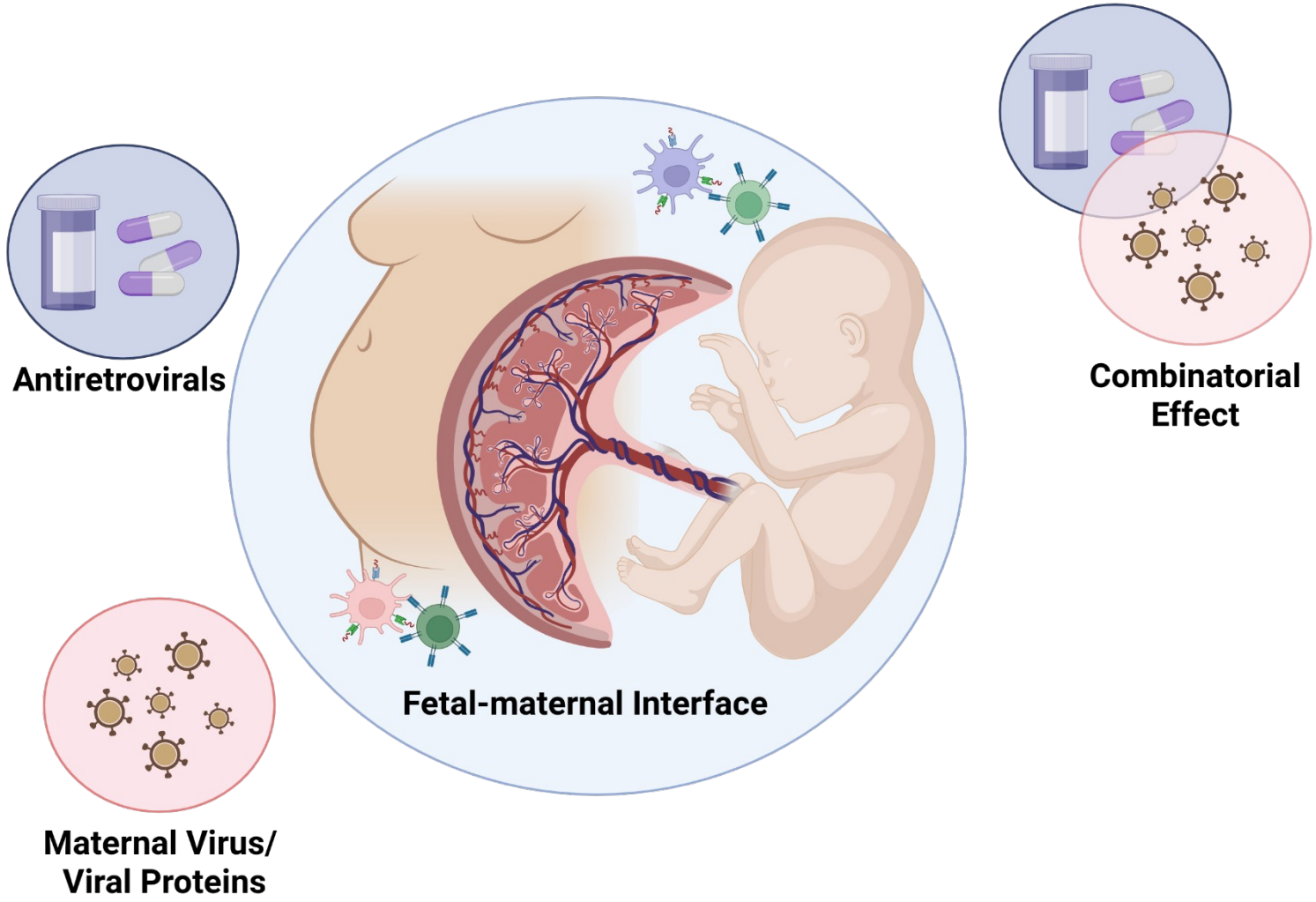
Full biological discovery potential often out of reach



- For **most** spectral flow cytometry datasets, only a **handful of cell populations** end up being analyzed by any method

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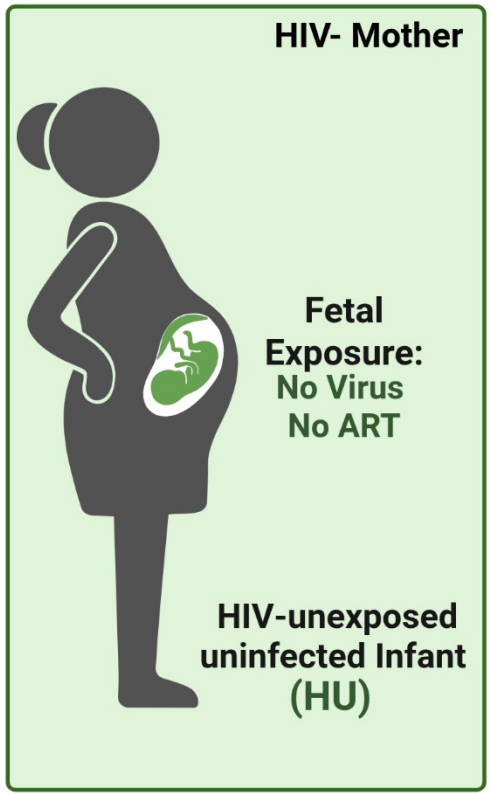
My PhD's focus: HIV-exposed Uninfected Infants

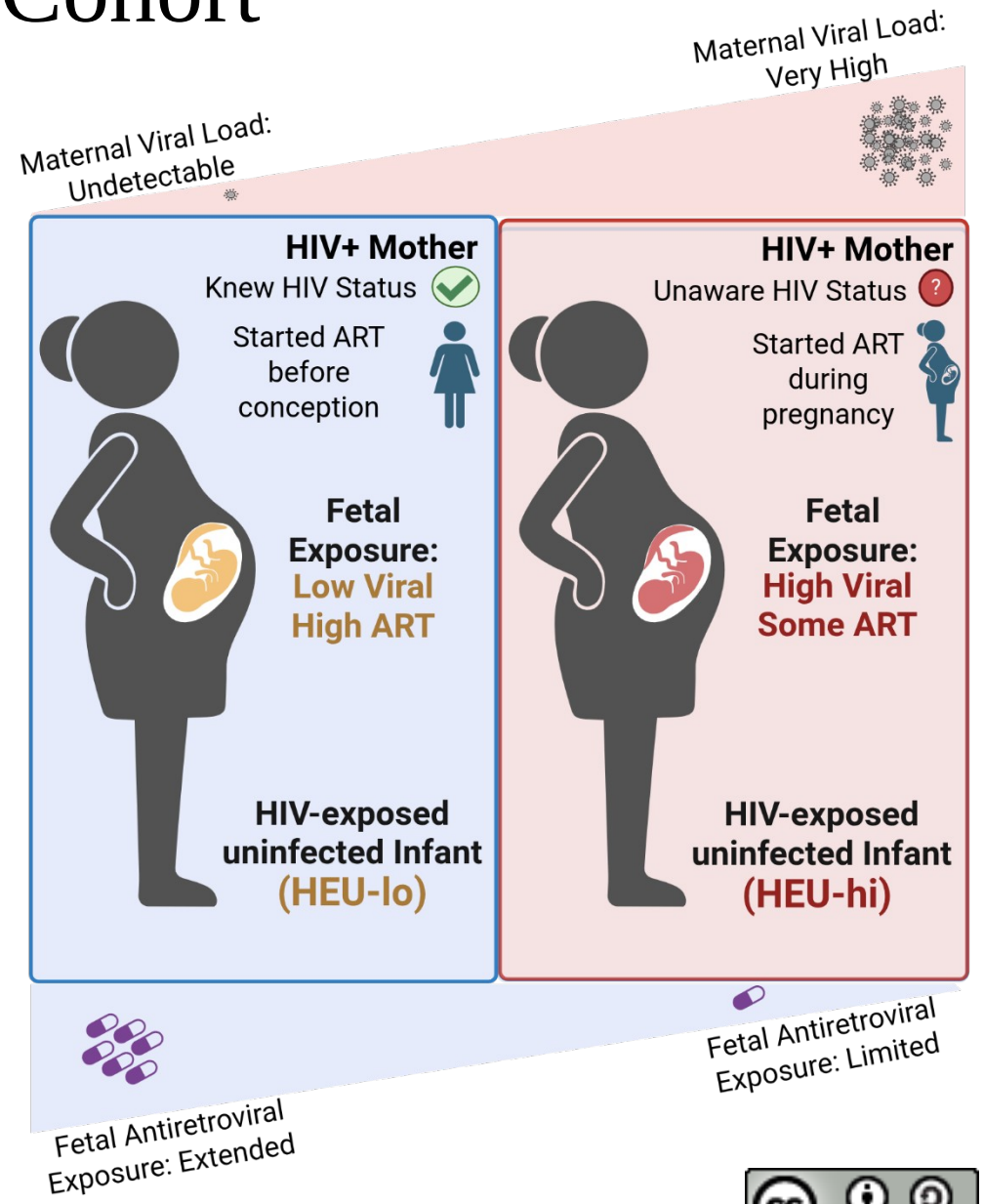


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Our Malawian Cohort



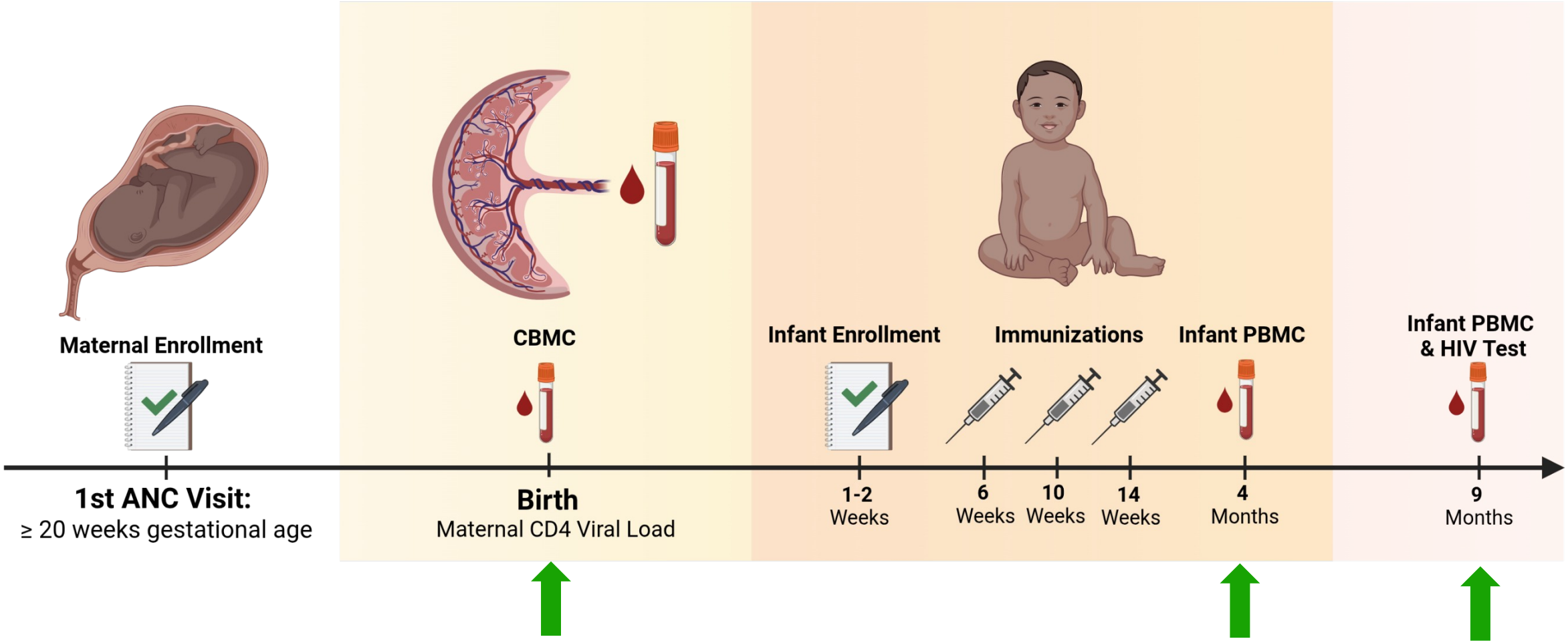




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Study Timeline



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32-Fluorophore Spectral Flow Cytometry Panel

UV1	Spark UV 387		
UV2	BUV395	CD62L	(SK11)
UV6	cFluor UV440		
UV7	BUV496		
UV9	BUV563	CD69	(FN50)
UV10	BUV615	CCR4	(1G1)
UV11	BUV661	Vδ2	(B6)
UV14	BUV737	CD38	(HB7)
UV16	BUV805	CD4	(SK3)
V1	BV421/		
V2	Spark Violet 423		
V3	Pacific Blue	CD14	(M5E2)
V3	Pacific Blue	CD19	(SJ25C)
V4	SYTOX Blue		
V5	BV480	CD161	(HP-3G10)
V6	SBV515		
V7	BV510		
V8	BV570	CD16	(3G8)
V9	Qdot 585		
V10	BV605	CD45RA	(HI100)
V11	BV650	CD8	(RPA-T8)
V13	BV711	Va7.2	(3C10)
V14	BV750	IFNγ	(B27)
V15	BV786	CCR6	(11A9)
V16	LD NIR 786		
B2	Alexa Fluor 488	FoxP3	(236A/E7)
B3	Spark Blue 550	CD3	(SK7)
B4	Spark Blue 574	CD45	(2D1)
B6	RB613	PD1	(EH12.1)
B7	NovaFluorBlue660		
B8	SBB675		
B9	NovaFluorBlue690		
B10	RB705	CD26	(M-A261)
B12	RB744/		
B13	NovaFluorBlue760		
B14	RB780	CXCR5	(RF8B2)
YG1	PE	ICOS	(ISA-3)
YG2	Spark YG 593		
YG3	PE-Dazzle 594	TNFα	(MAB11)
YG4	PE-Fire 640/		
YG5	PE-Cy5	CXCR3	(G025H7)
YG6	NovaFluorYellow690		
YG7	PE-Fire 700	CD127	(A019D5)
YG8	PE-Fire 744	CD25	(M-A251)
YG9	PE-Vio 770	HLA-DR	(REA805)
YG10	PE-Fire810		
R1	APC	CD39	(A1)
R2	Alexa Fluor 647	IL2	(MQ1-
R3	cFluor R685		
R4	APC-R700	CD107a	(H4A3)
R5	NovaFluorRed725		
R6	Zombie NIR	L/D	
R7	APC-Fire 750	CD27	(O323)
R8	APC-Fire 810	CCR7	(G043H7)

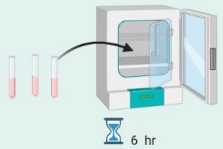
Thaw & Rest



1) Thaw



2) Count

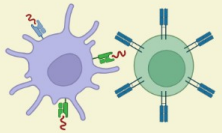


3) Rest

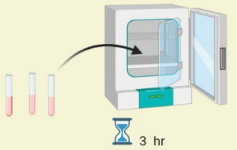
Antigen Incubation



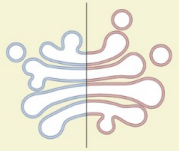
4) Aliquot by condition



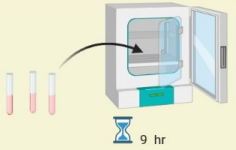
5) Add antigens (PPD, SEB, Ctrl)



6) Incubate



7) Add Golgi Transport Inhibitors



8) Incubate

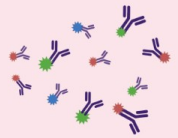
Flow Cytometry Staining



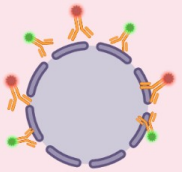
9) Pellet



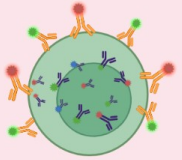
10) Viability stain



11) Surface antibodies



12) Fix & permeabilize



13) Intracellular antibodies



14) Resuspend in 0.4% PFA



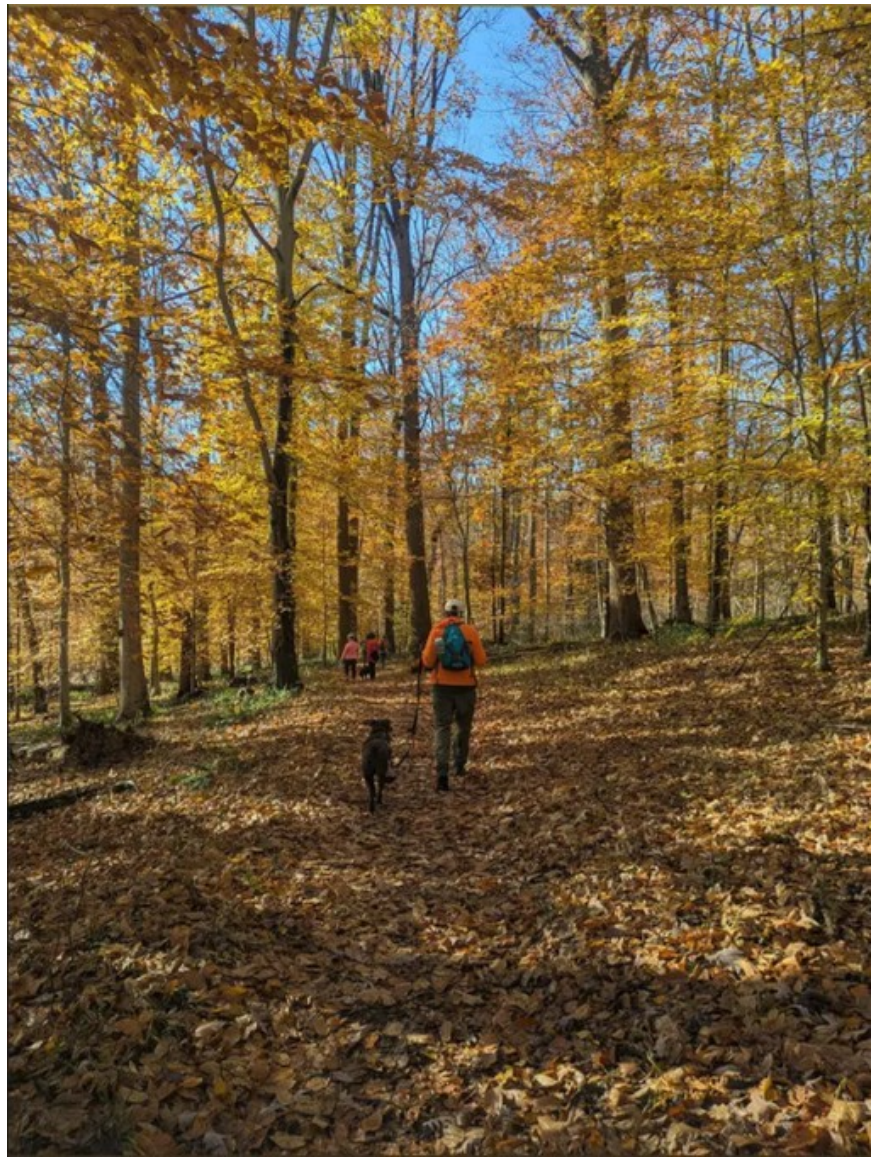
15) Refrigerate at 4*C



16) Acquisition Cytek Aurora 5L

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Personal Goal: No cell left behind



Repeated 3 AM thoughts

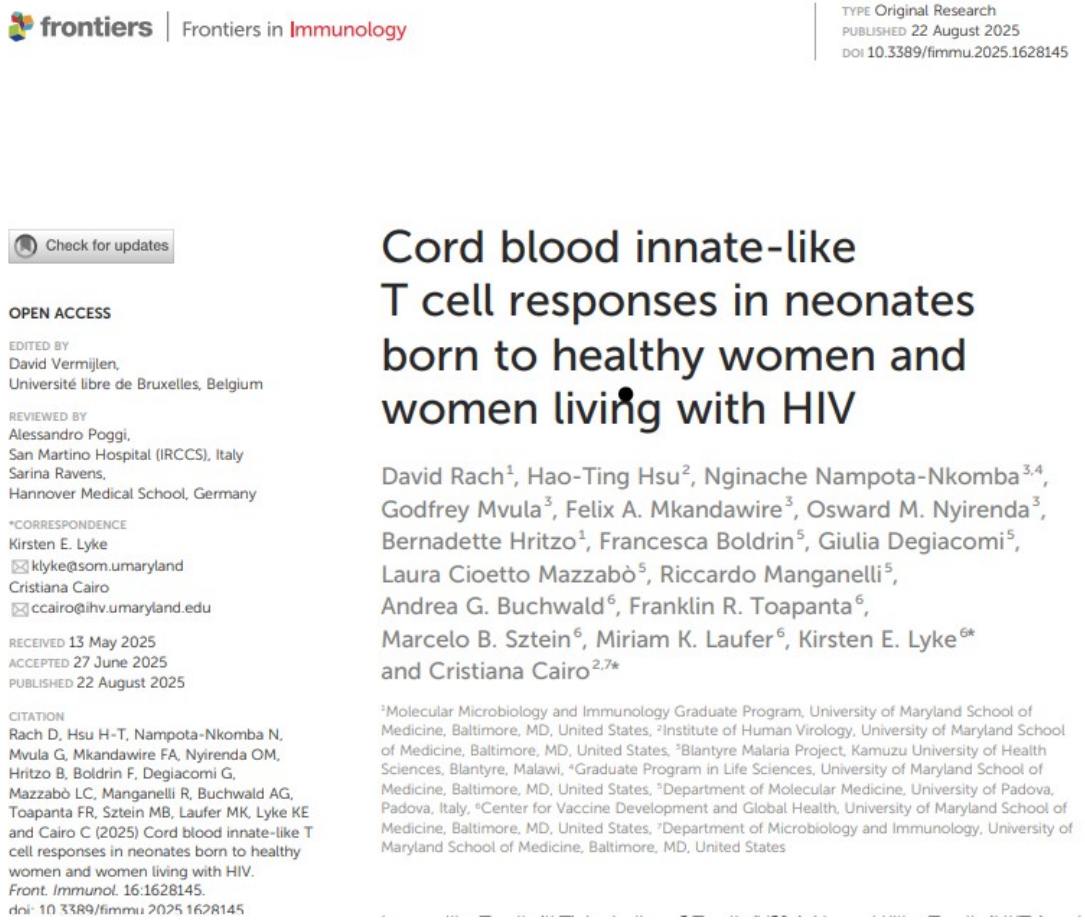
- ❑ What makes spectral flow cytometry data amazing is both the depth and the breadth

- ❑ How to fully profile the entirety of the acquired dataset, not just a single cell population of interest

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Previously looked at unconventional T cells

Paper



frontiers | Frontiers in Immunology

TYPE Original Research
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Check for updates

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Cord blood innate-like T cell responses in neonates born to healthy women and women living with HIV

David Rach[1], Hao-Ting Hsu[2], Nginache Nampota-Nkomba[3,4], Godfrey Mvula[3], Felix A. Mkandawire[3], Osward M. Nyirenda[3], Bernadette Hritzo[1], Francesca Boldrin[5], Giulia Degiacomi[5], Laura Cioetto Mazzabò[5], Riccardo Manganelli[5], Andrea G. Buchwald[6], Franklin R. Toapanta[6], Marcelo B. Sztein[6], Miriam K. Laufer[6], Kirsten E. Lyke[6*] and Cristiana Cairo[2,7*]

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Main differences this time around:

- ☐ Shifting from 50%-50% FlowJo – R to increasingly more R
- ☐ Targeted re-writes in Rust to speed up original R code
- ☐ Building around reproducibility from the start (not as an after thought)
- ☐ Improved Bioconductor interoperability

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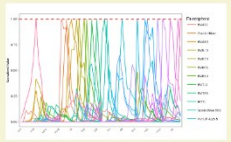
Semi-supervised Pipeline



1) Acquisition
(Cytek Aurora 5L)



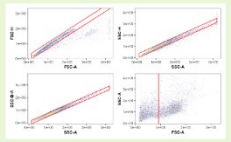
2) Unmixing Controls
(Luciernaga)



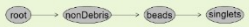
3) Unmixing
(OLS, TRU-OLS, AutoSpectral)



4) File Checks
(PeacoQC, Flow AI, etc)



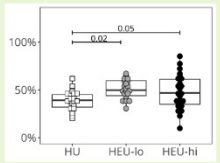
5) Clean Up Gates
(FlowJo or flowWorkspace)



6) Initial Gates
(openCyto or flowMagic)



7) FlowJo .wsp
(CytoML Docker)



8) Major subset Outputs

Iterating by Experiment



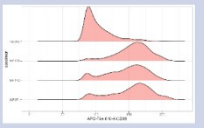
9) Subset Export



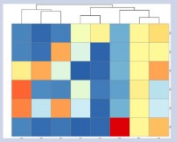
10) Return to R
(flowWorkspace)



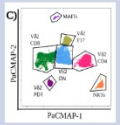
11) Coereba



12) Normalization
(CytoNorm or CyCombine)



13) Clustering
(FlowSOM)



14) Dimensionality Visualization
(UMAP or PaCMAP)



Summarized Experiment



15) Coereba

Iterating by Cell Subset

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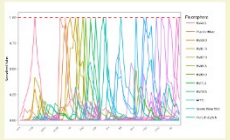
Pipeline: Acquisition



1) Acquisition
(Cytek Aurora 5L)



2) Unmixing Controls
(Luciernaga)



3) Unmixing
(OLS, TRU-OLS, AutoSpectral)



4) File Checks
(PeacoQC, Flow AI, etc)

Iterating by Experiment

- Acquired day of staining on a 5-Laser Cytek Aurora
- Instrument flushed with 10% Contrad and Water for 20 minutes pre-use
- Instrument passed by both the UMGCCC FCSR and Cytek's QC criteria immediately before acquisition

UMGCCC FCSR InstrumentQC

Plots for our 5L Aurora

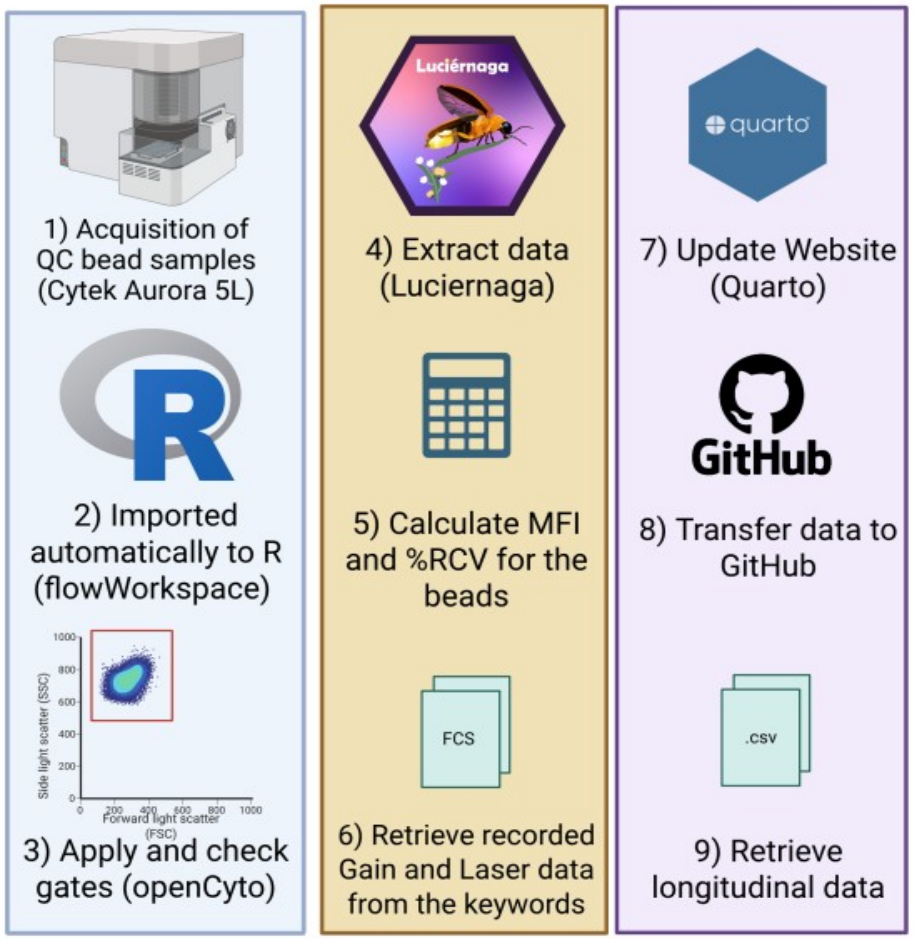
Historical Data 5L Aurora

Dashboard How-To-Set-Up

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Automated Daily Processing





Behind-the-scenes:

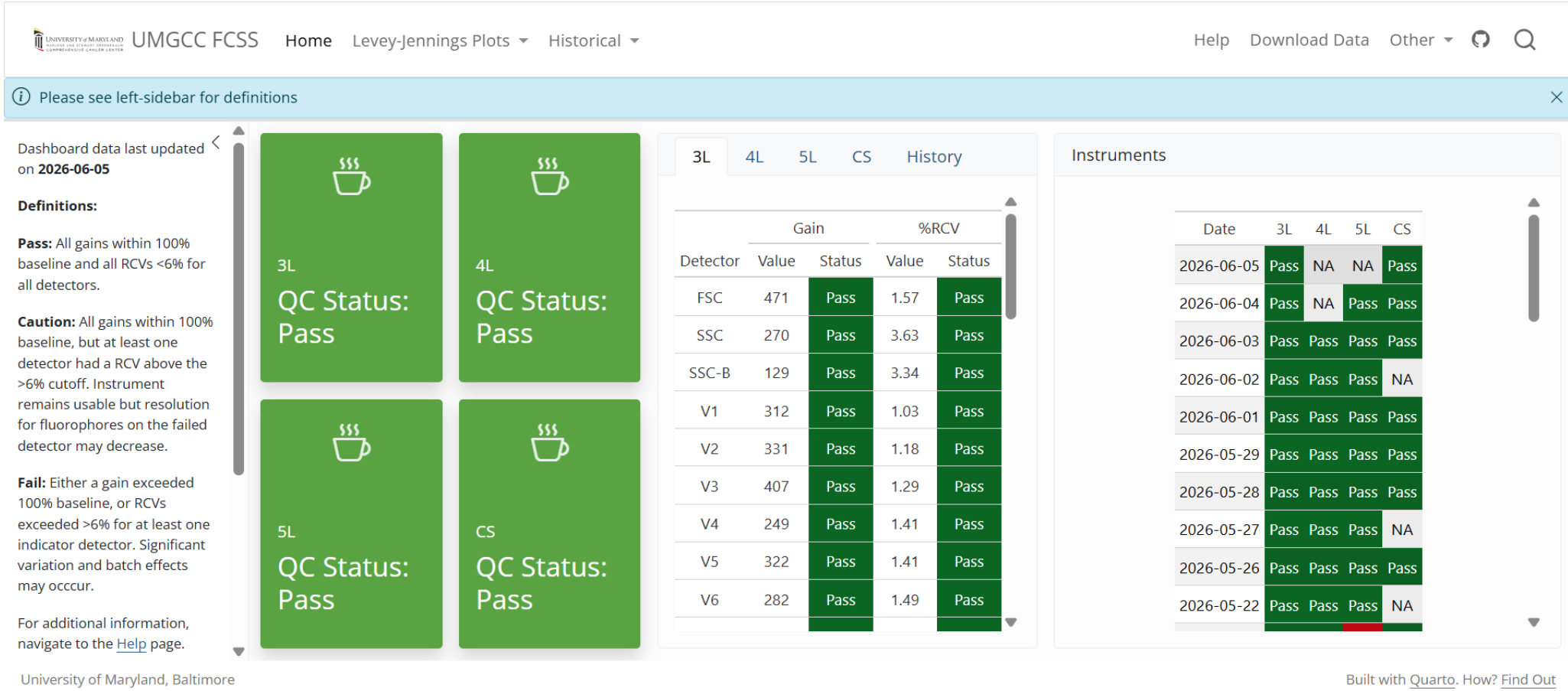
- Using R (and increasingly Rust) to process the .fcs files
- Automated transfer to GitHub via Windows Task Scheduler
- Our Quarto Website gets updated twice daily via a GitHub action

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Instrument Quality Control Dashboard

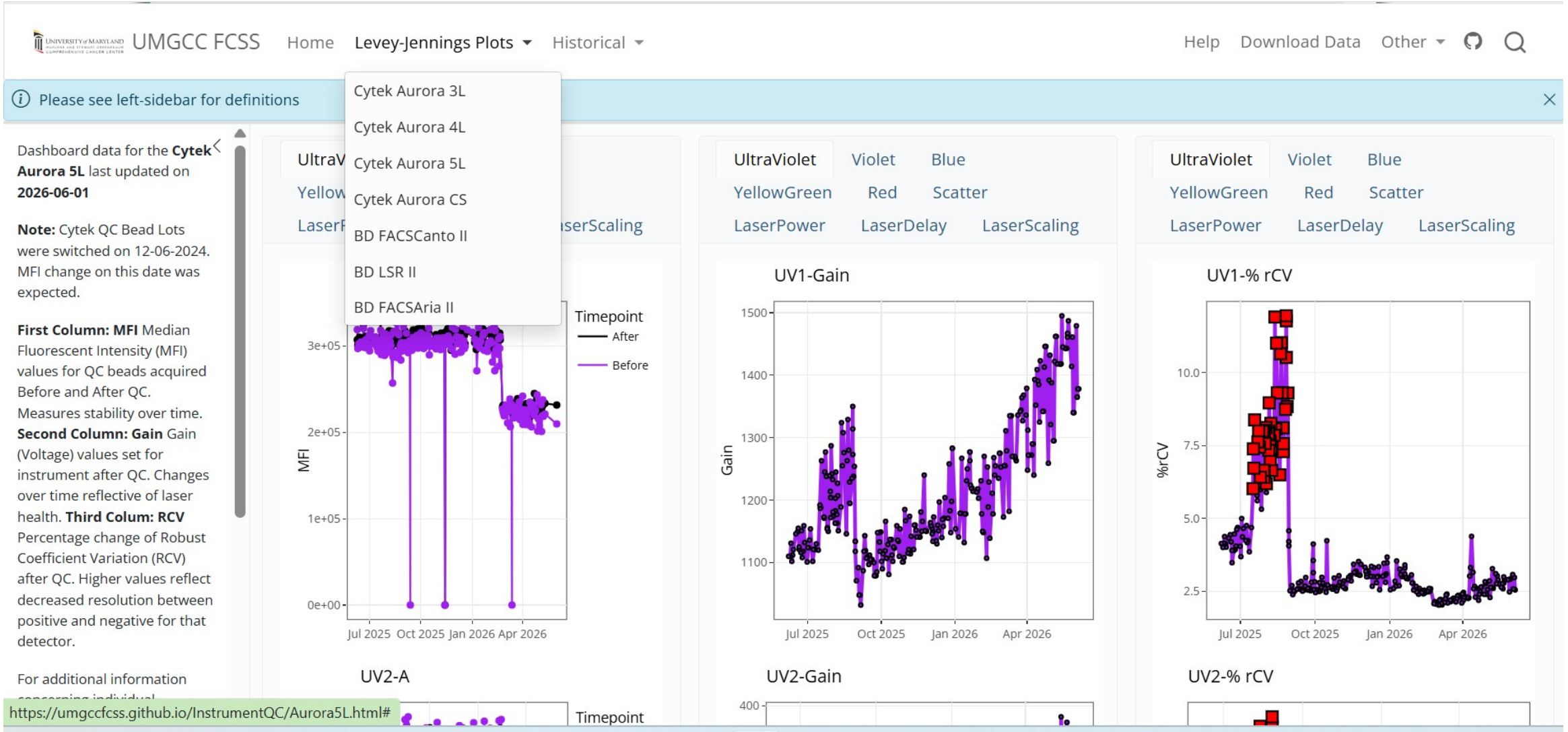
Dashboard

How-To

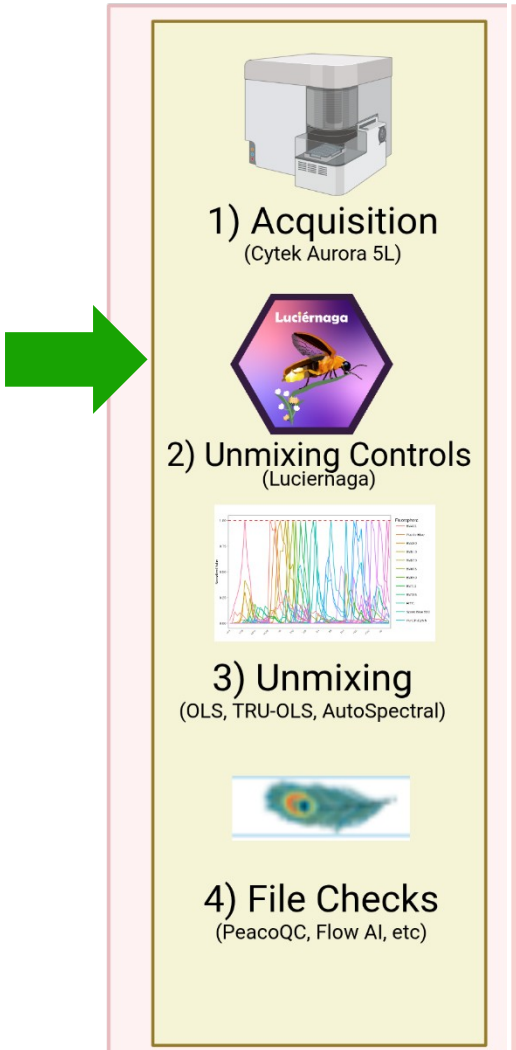


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Instrument Quality Control Dashboard



Pipeline: Evaluating Unmixing Controls



- ❖ New unmixing controls acquired with each experiment.
- ❖ Extra specimen cells acquired as additional unstained controls

- ❑ Unmixing Controls are screened for suitability for unmixing
- ❑ Single-colors: Tandem degradation, and autofluorescence contamination
- ❑ Unstained(s): Presence of multiple-autofluorescences in lymphocyte gate

- ❑ Flagged unmixing controls are replaced with equivalents from adjacent experimental date

Luciernaga

Luciernaga Vignette

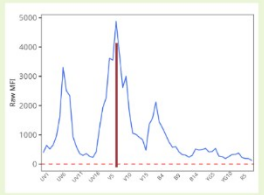
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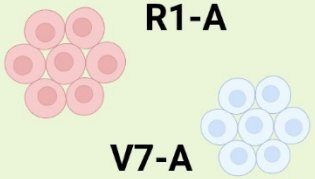
Luciernaga: Individual Cell Normalized Signatures



Code



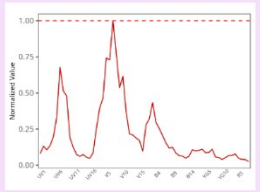
Identify detector with brightest MFI for each cell



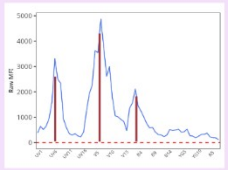
Group cells with shared peak detectors



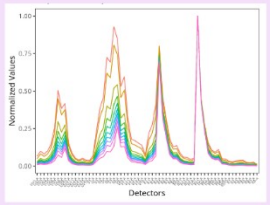
Subtract Out Autofluorescence



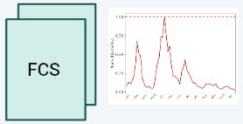
Normalize individual cell fluorescent signatures



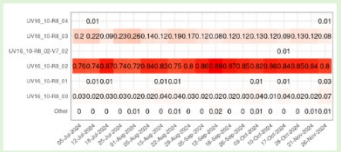
Identify peaks using local maxima



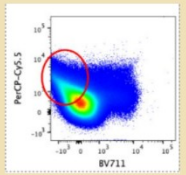
Group cells with shared peaks & heights



Enumerate and export out cells that share signature as .fcs files



Checks for tandem degradation

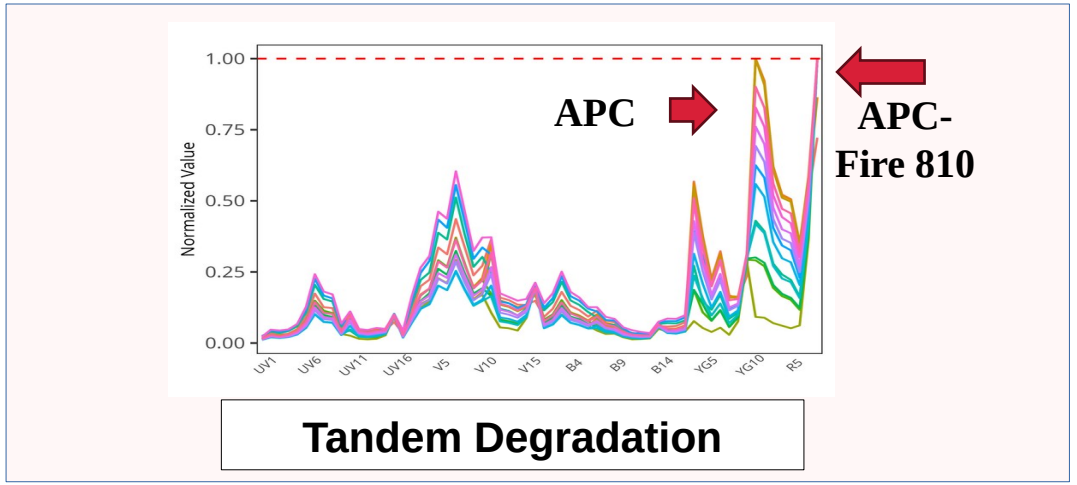


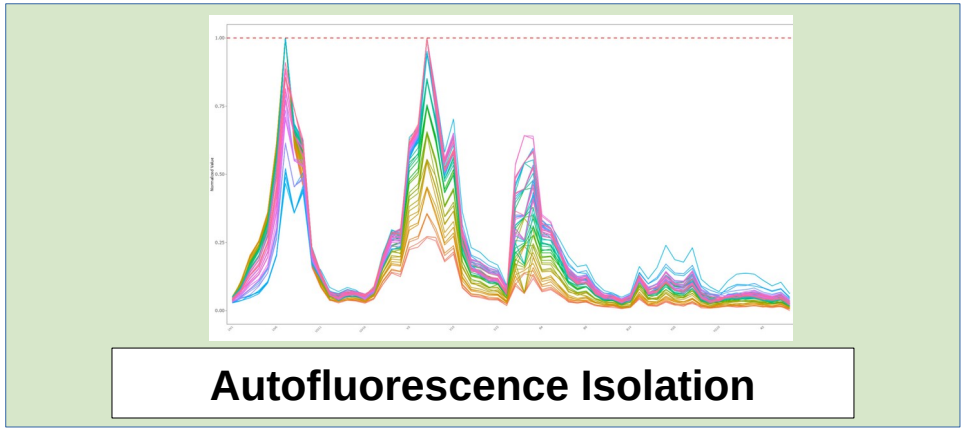
Autofluorescence Isolation

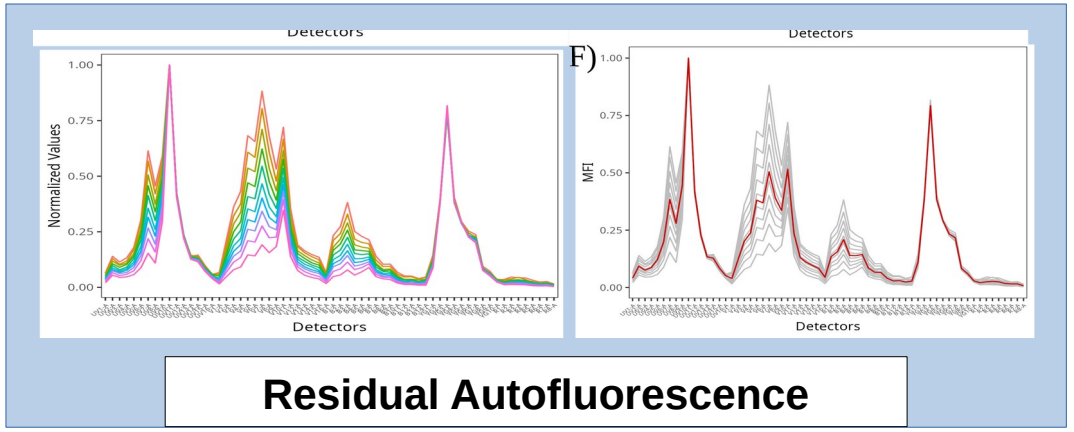
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Automated tools for screening unmixing controls







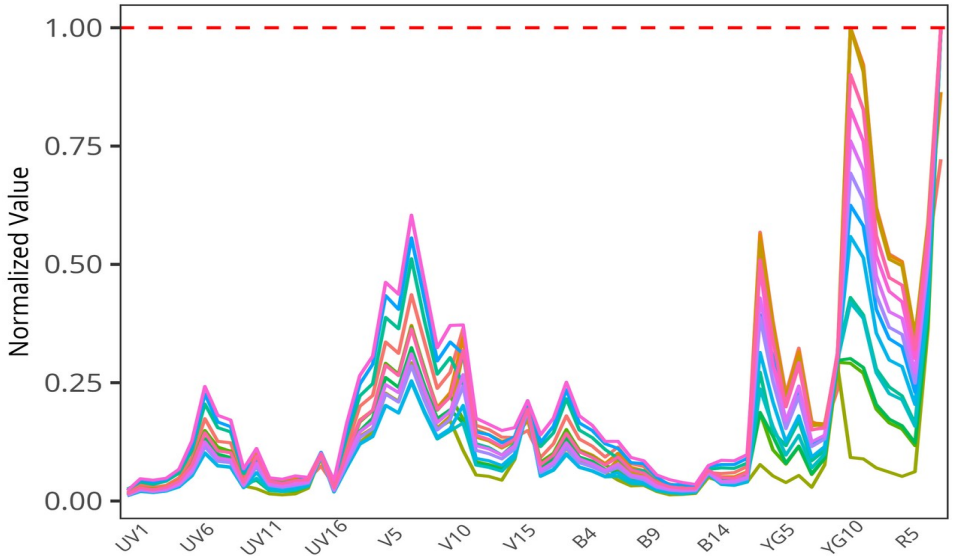
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Identifying degrading fluorophores

Fluorophores in good shape → **one predominant normalized signature** within the unmixing control raw .fcs file

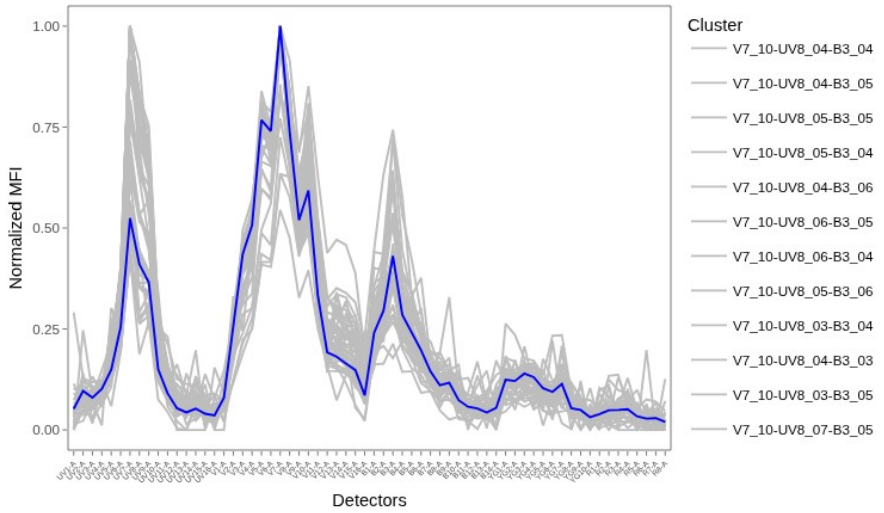
- When the fluorophore degrades, noticeable **increase in variant signatures**, making up substantial proportion of all cells in the unmixing control

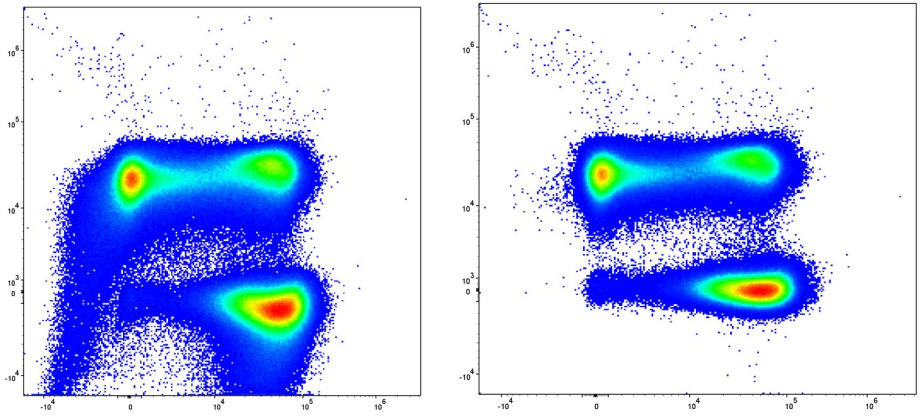
- Automated screening all acquired unmixing controls from campus users during off-hours



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Isolating additional autofluorescence signatures





- This approach also works to isolate variant autofluorescence signatures from unstained controls

- Unlike other approaches, the isolated cells share normalized signatures, only differing in terms of brightness

- Return directly as an .fcs file containing just cells with same signature for use as an autofluorescent tag in SpectroFlo

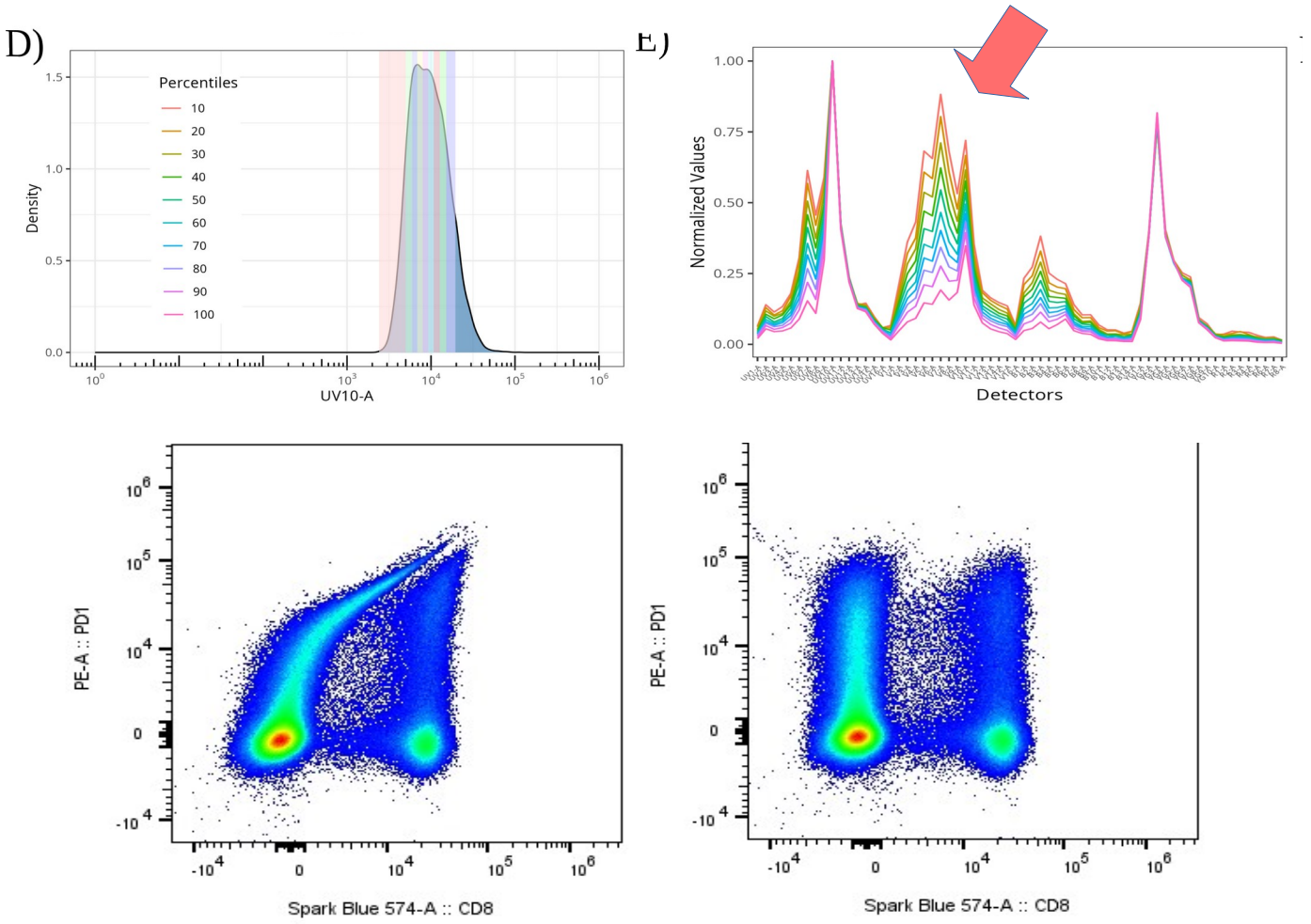
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Residual autofluorescence in dim cell single-color controls

- **“As bright or brighter”**

- Dim antigen-fluorophore combinations acquired as **cell single-color controls** often have residual autofluorescence contribution to their signature

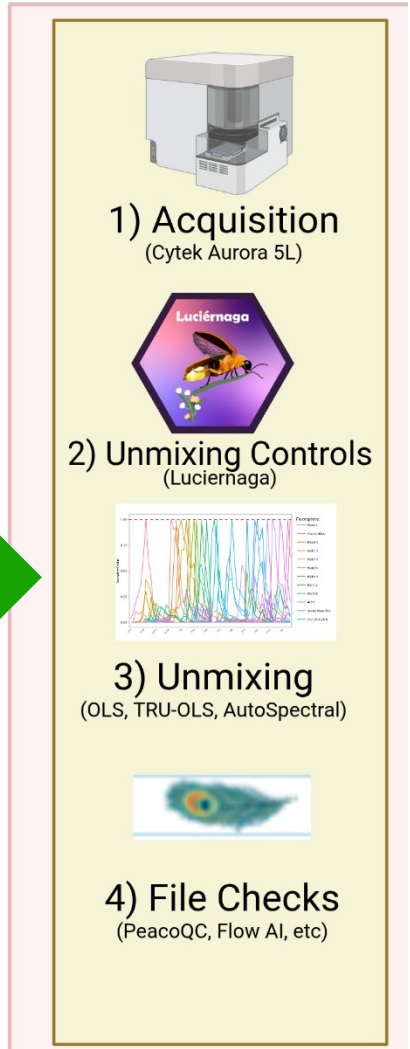
- Identifying and **switching these** dim cell single-color controls to **bead single-color controls** significantly improved unmixing for our panels



Signature with residual AF

Signature without residual AF

Pipeline: Unmixing



- ❖ Unmixing Method Agnostic
- ❑ Unmixing in SpectroFlo (with modified single-color and unstained inputs via Luciernaga)
- ❑ Unmixing in R
  - OLS (Luciernaga using Rust)
  - TRU-OLS (via Julia wrapper)
  - AutoSpectral

TRU-OLS

TRU-OLS Walkthrough

AutoSpectral

AutoSpectral Walk-through

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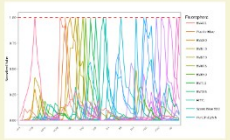
Pipeline: File Checks



1) Acquisition
(Cytek Aurora 5L)



2) Unmixing Controls
(Luciernaga)



3) Unmixing
(OLS, TRU-OLS, AutoSpectral)



4) File Checks
(PeacoQC, Flow AI, etc)

Iterating by Experiment

- ❑ Checking for fluidic or laser-related issues in the unmixed files
- ➢ R: PeacoQC or FlowAI
- ➢ Rust: PeacoQC (thanks James!)

PeacoQC

PeacoQC Vignette

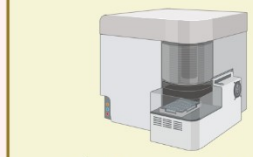
FlowAI

FlowAI Vignette

PeacoQC in Rust

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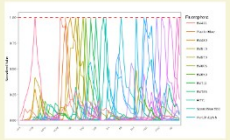
Pipeline: Manual and Automated Gating



1) Acquisition
(Cytek Aurora 5L)



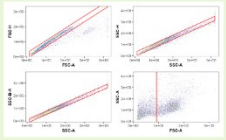
2) Unmixing Controls
(Luciernaga)



3) Unmixing
(OLS, TRU-OLS, AutoSpectral)



4) File Checks
(PeacoQC, Flow AI, etc)



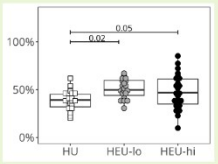
5) Clean Up Gates
(FlowJo or flowWorkspace)



6) Initial Gates
(openCyto or flowMagic)



7) FlowJo .wsp
(CytoML Docker)



8) Major subset Outputs

Iterating by Experiment

- ❑ Doublets, Debris, Dead cell cleanup (in R or FlowJo)
- ❑ Automated Gating for major cell subsets (openCyto or flowMagic)
- ❑ All automated gate placements visualized via export to a .pdf

flowGate

flowGate Walkthrough

openCyto

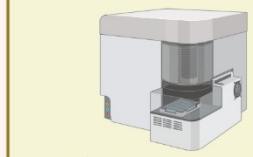
openCyto Walk-through

flowMagic

flowMagic

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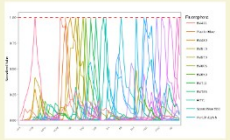
Pipeline: Save to FlowJo v10 .wsp



1) Acquisition
(Cytek Aurora 5L)



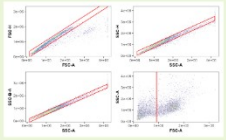
2) Unmixing Controls
(Luciernaga)



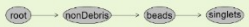
3) Unmixing
(OLS, TRU-OLS, AutoSpectral)



4) File Checks
(PeacoQC, Flow AI, etc)



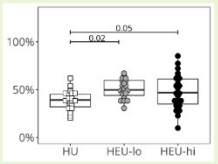
5) Clean Up Gates
(FlowJo or flowWorkspace)



6) Initial Gates
(openCyto or flowMagic)



7) FlowJo .wsp
(CytoML Docker)



8) Major subset Outputs

Iterating by Experiment

- ❑ Everything is saved to a FlowJo v10 .wsp, via CytoML
- ❑ Any automated gates that need adjustment get corrected
- ❑ Major cell subset statistics exported and analyzed

flowWorkspace

CytoML

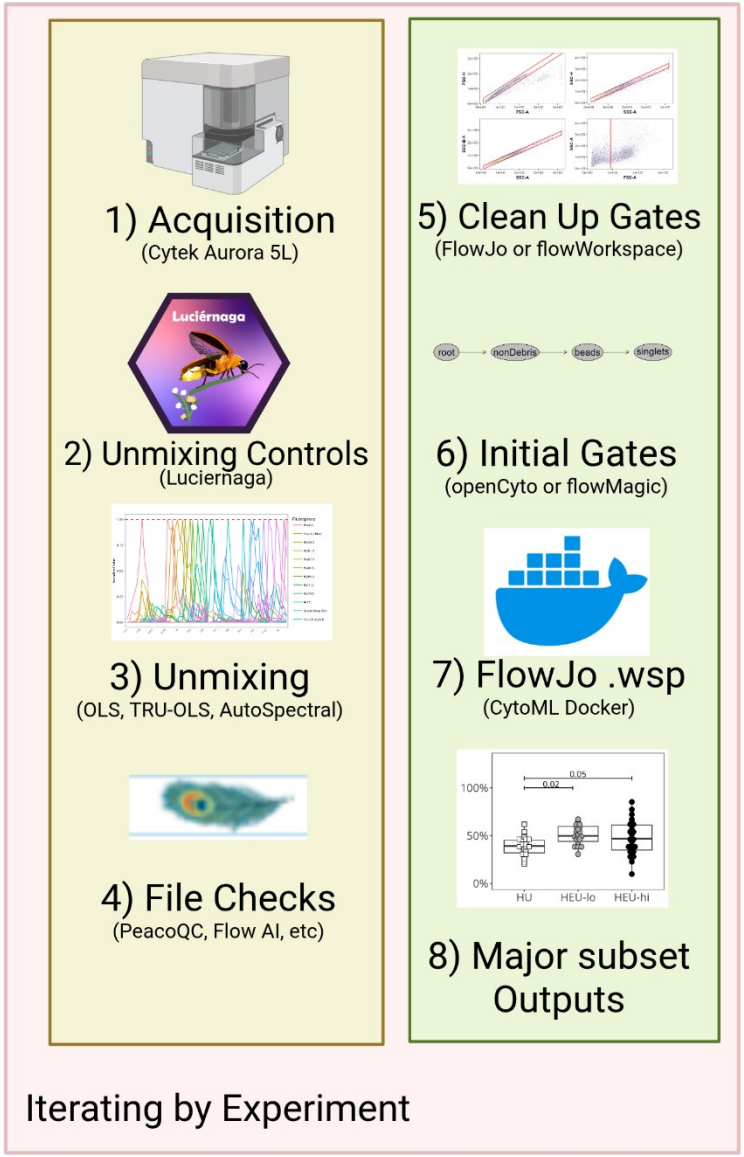
FlowJo to R

R to FlowJo

Data for Statistics Walkthrough

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Pipeline: Iterating through cell subsets



Subset Export Walkthrough

GatingSets Walkthrough

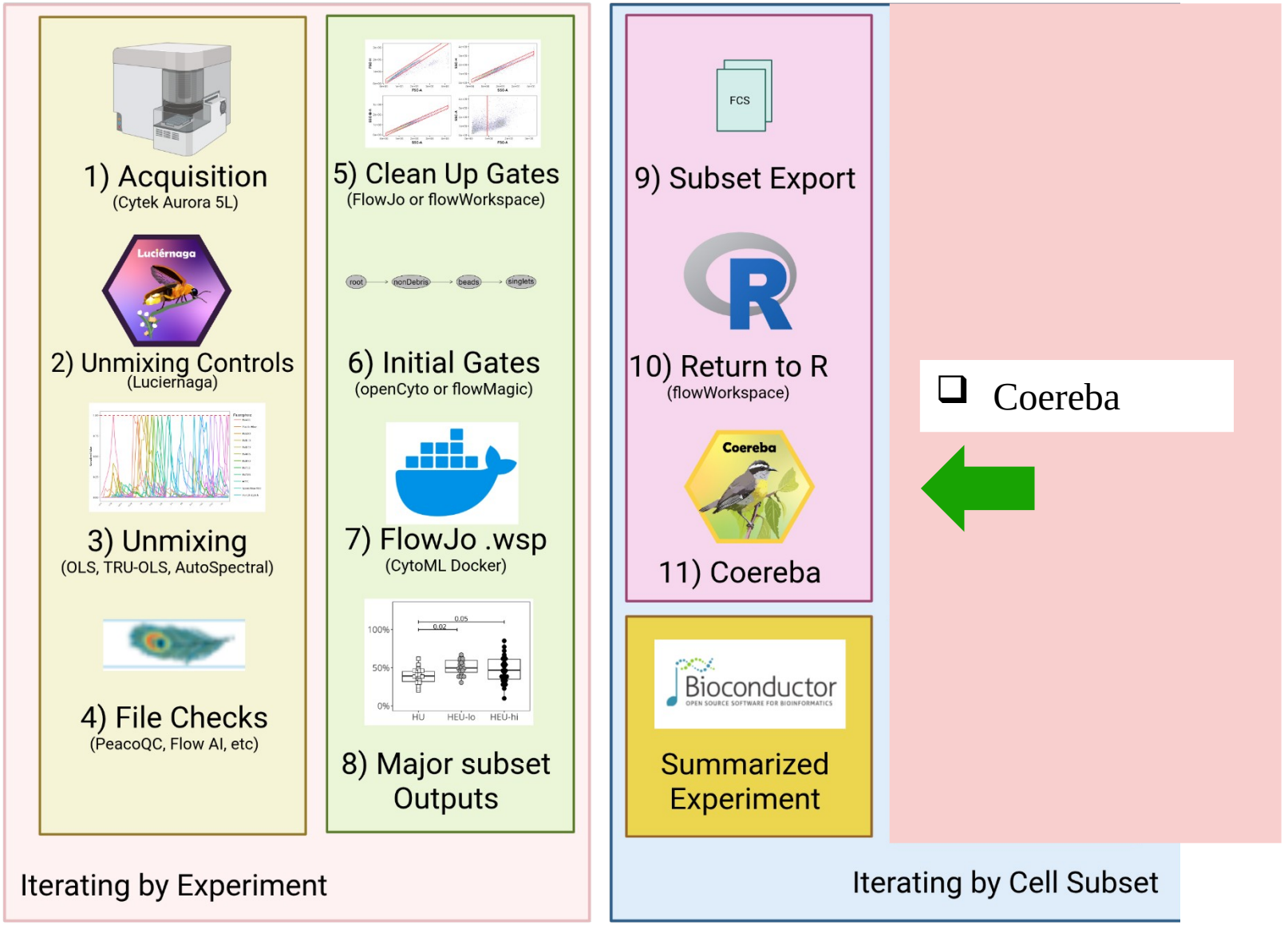
Coereba

Coereba Vignette

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University of Maryland School of Medicine

Pipeline: Coereba



Subset Export Walkthrough

GatingSets Walkthrough

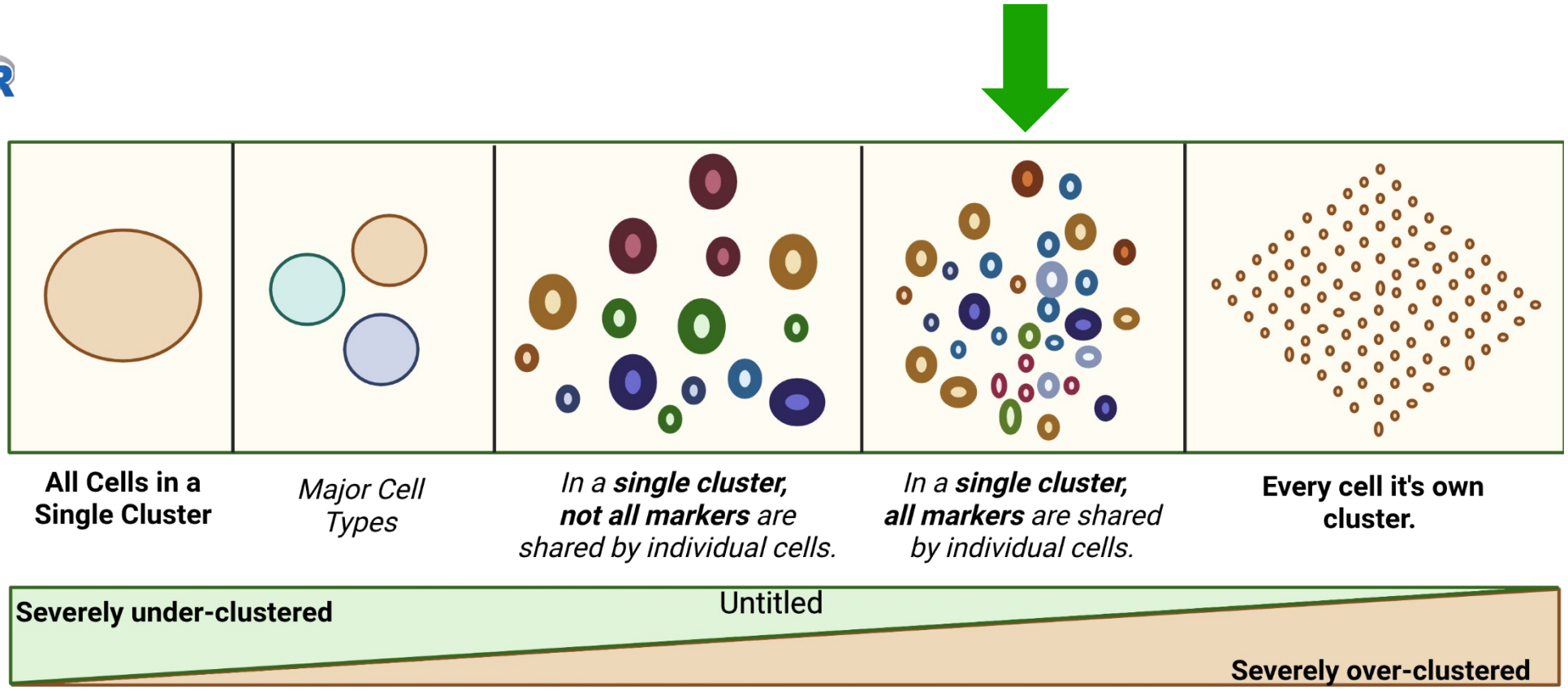
Coereba

Coereba Vignette

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Grouping cells that share every marker



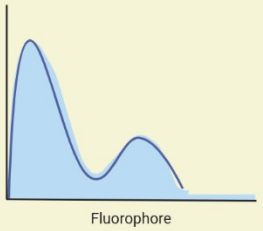
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Coereba

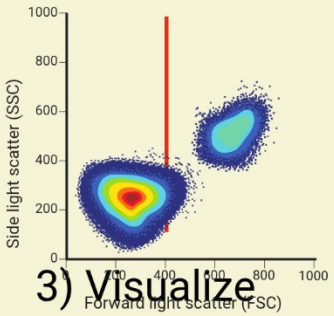


Code

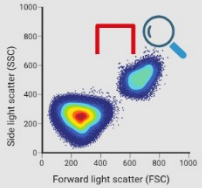
1) Load into R
(flowWorkspace)



2) Calculate Splitpoints

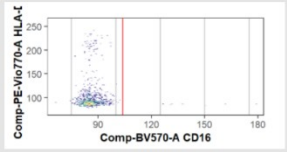


3) Visualize

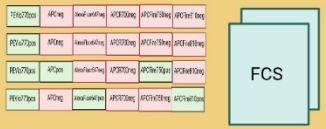


4) Gate Constraints

5) Shiny App adjustments



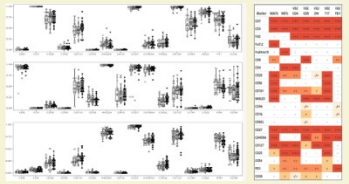
6) Splitpoints updated



7) CoerebaID

Bioconductor
OPEN SOURCE SOFTWARE FOR BIOINFORMATICS

8) Summarized Experiment

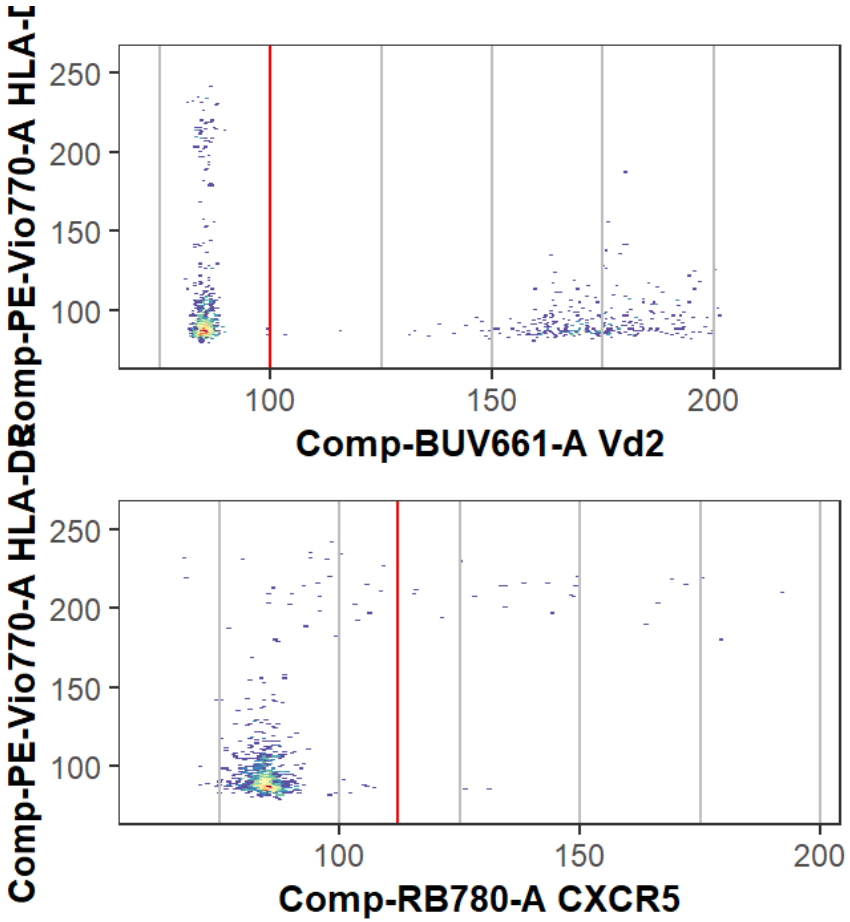


9) Visualize

University of Maryland
School of Medicine

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Initial automated gate, with ability to adjust

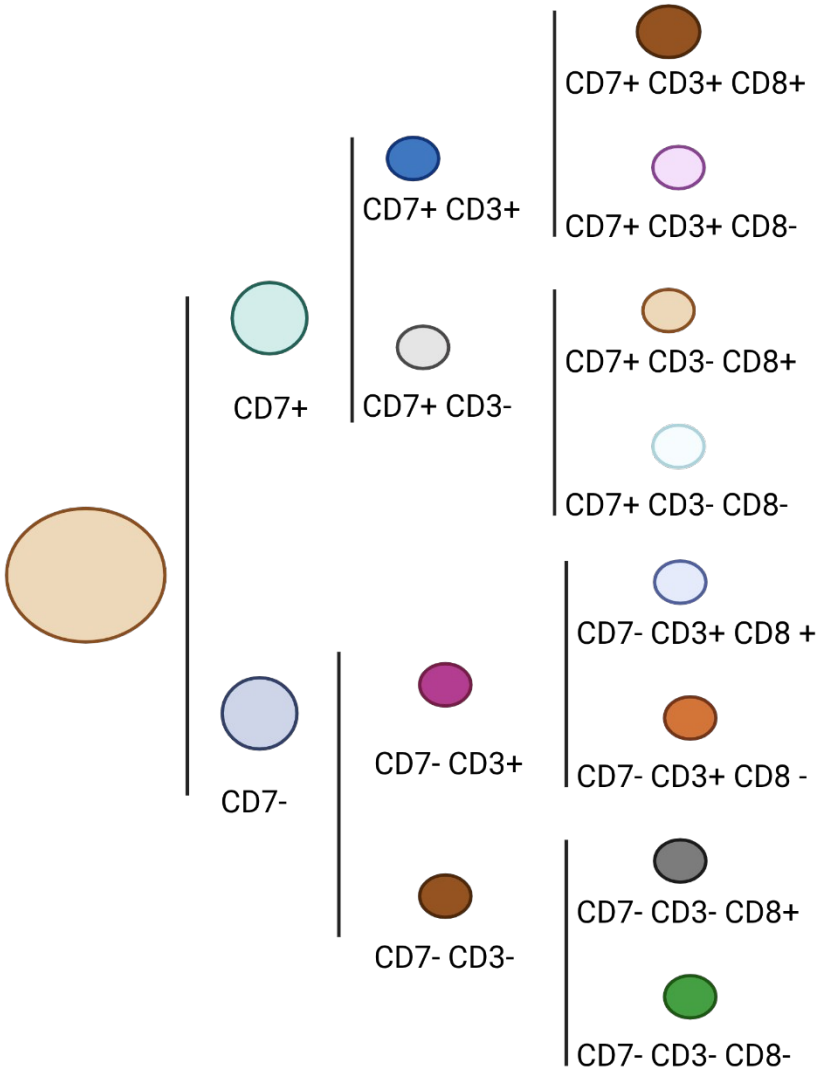


- ❑ Initial gates for designated markers calculated via openCyto (mindensity)
- ❑ Ability to update gate_range constraints
- ❑ Launch a Shiny app and designate interactively

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CoerebaID column gets added to the .fcs file

PEVio770pos	APCneg	AlexaFluor647neg	APCR700neg	APCFire750neg	APCFire810neg
PEVio770pos	APCneg	AlexaFluor647neg	APCR700neg	APCFire750neg	APCFire810neg
PEVio770pos	APCpos	AlexaFluor647neg	APCR700neg	APCFire750pos	APCFire810neg
PEVio770pos	APCneg	AlexaFluor647pos	APCR700neg	APCFire750neg	APCFire810pos



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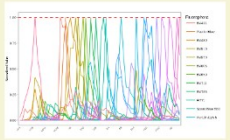
Pipeline: Summarized Experiment



1) Acquisition
(Cytek Aurora 5L)



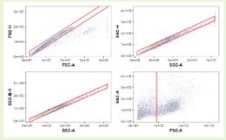
2) Unmixing Controls
(Luciernaga)



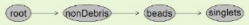
3) Unmixing
(OLS, TRU-OLS, AutoSpectral)



4) File Checks
(PeacoQC, Flow AI, etc)



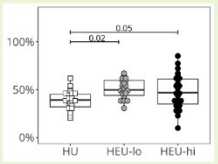
5) Clean Up Gates
(FlowJo or flowWorkspace)



6) Initial Gates
(openCyto or flowMagic)



7) FlowJo .wsp
(CytoML Docker)



8) Major subset Outputs

Iterating by Experiment



9) Subset Export



10) Return to R
(flowWorkspace)



11) Coereba



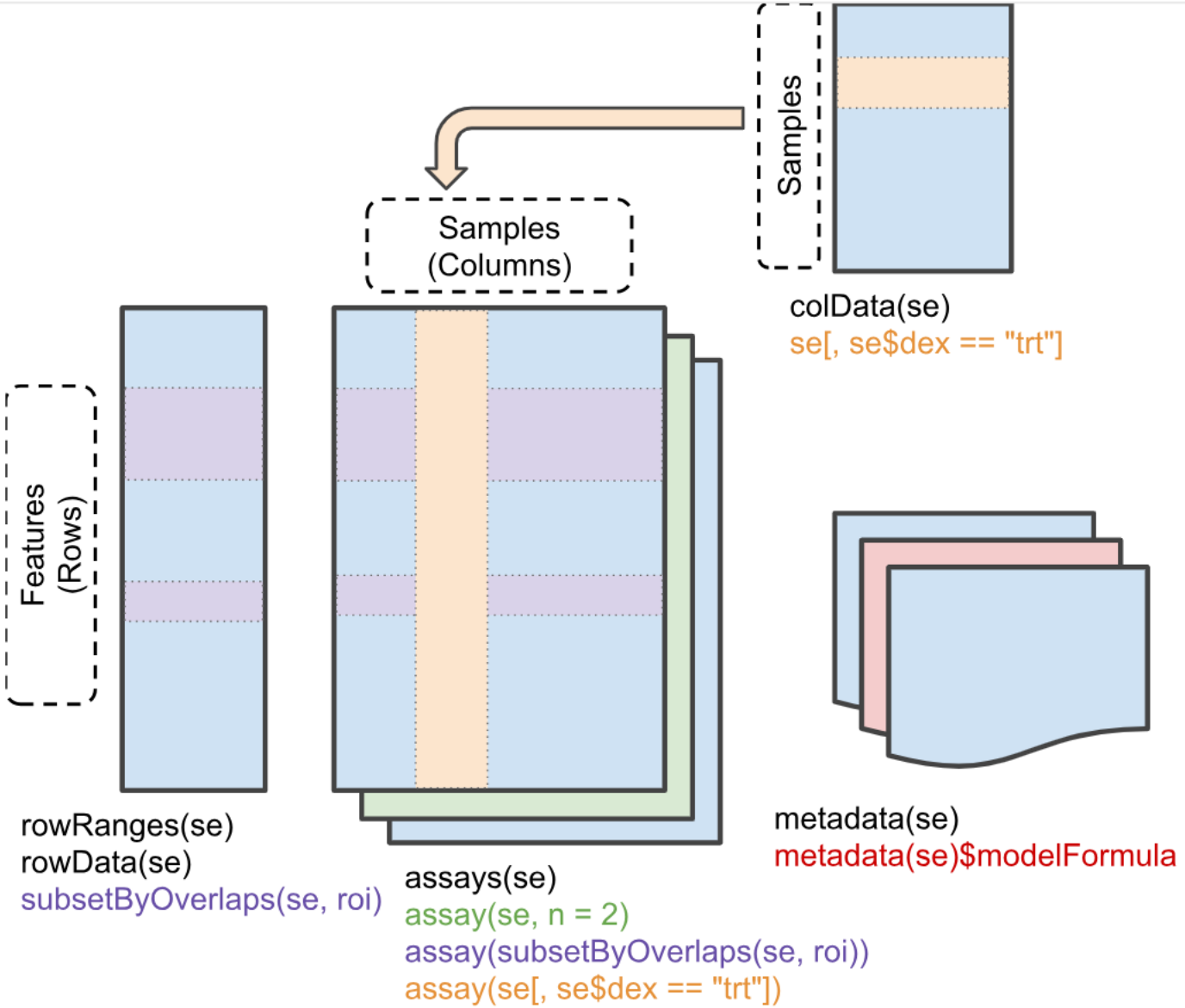
Summarized Experiment

Ite

- Data stored in a Summarized Experiment
- Interoperability with Bioconductor statistics packages

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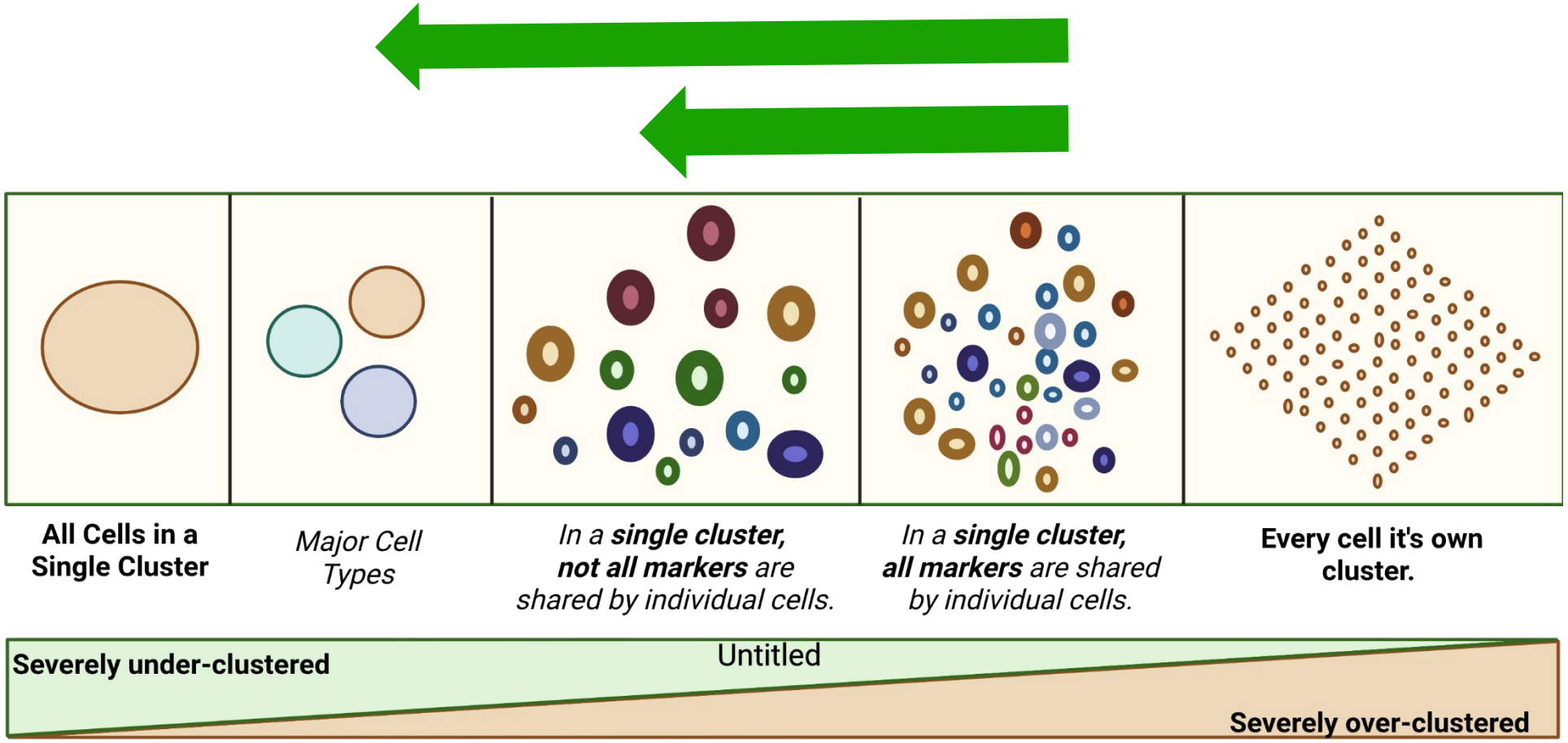
Summarized Experiment



- ❑ Permits rapid re-aggregation of the terminal nodes for re-evaluation
- ❑ Ties in with downstream Bioconductor statistics packages
- ❑ Portable. Don't like how we aggregated groups of cells? File is small enough to store in the same data repository as .fcs files for subsequent re-analysis.
- ❑ Filterable by metadata or node.
  - Nodes only in HEU-hi?
  - Nodes unique to one individual?

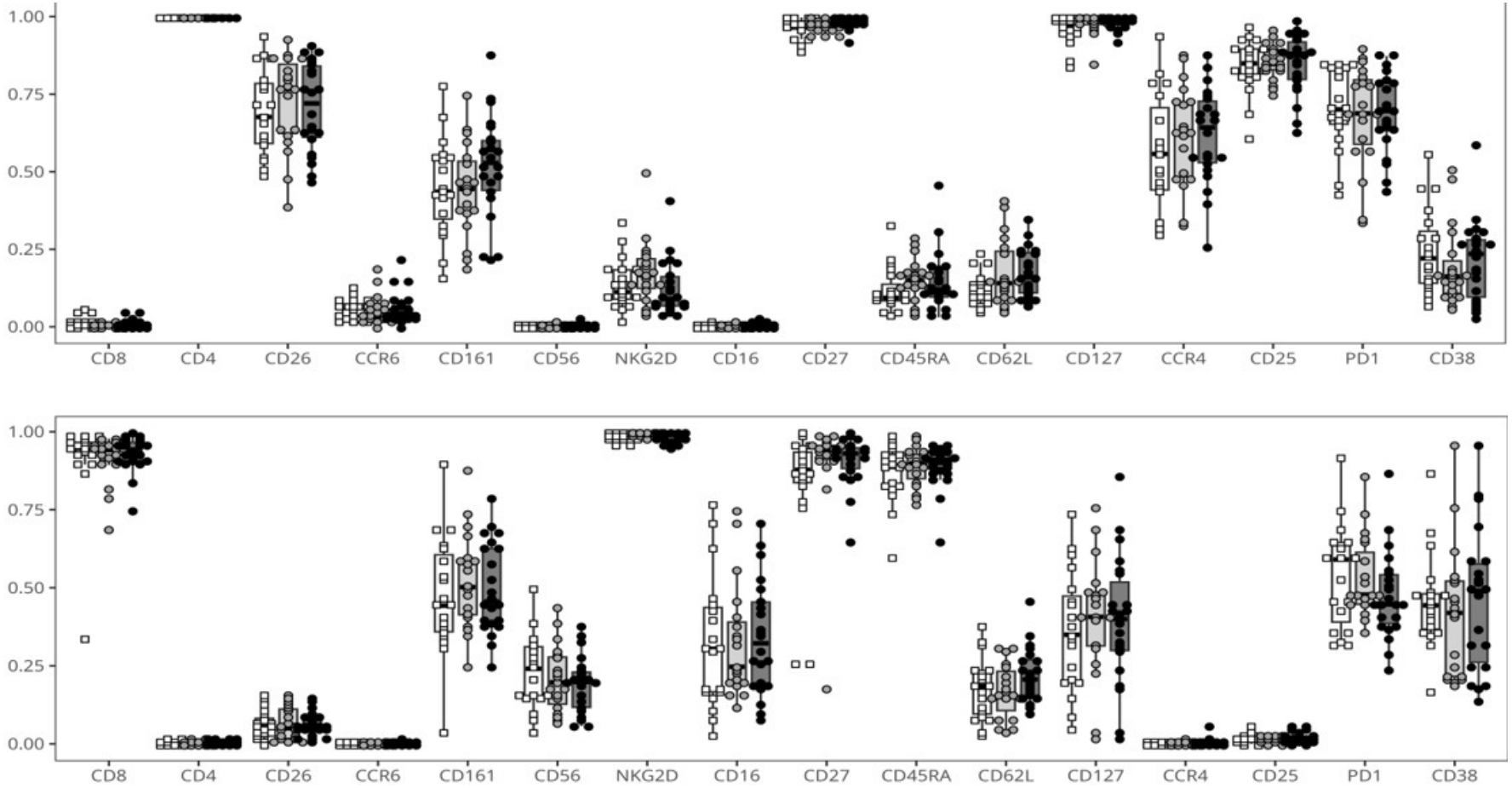
CC BY SA

Selectively re-aggregate in seconds



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Retrieval of marker expressions



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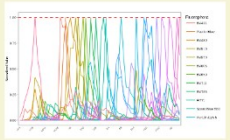
Pipeline: Normalization



1) Acquisition
(Cytek Aurora 5L)



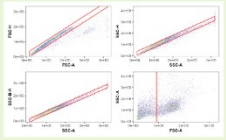
2) Unmixing Controls
(Luciernaga)



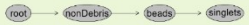
3) Unmixing
(OLS, TRU-OLS, AutoSpectral)



4) File Checks
(PeacoQC, Flow AI, etc)



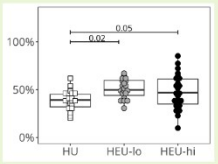
5) Clean Up Gates
(FlowJo or flowWorkspace)



6) Initial Gates
(openCyto or flowMagic)



7) FlowJo .wsp
(CytoML Docker)



8) Major subset Outputs

Iterating by Experiment



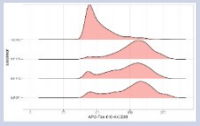
9) Subset Export



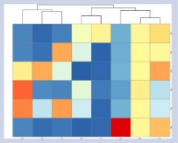
10) Return to R
(flowWorkspace)



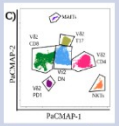
11) Coereba



12) Normalization
(CytoNorm or CyCombine)



13) Clustering
(FlowSOM)



14) Dimensionality Visualization
(UMAP or PaCMAP)



Summarized Experiment



15) Coereba

Iterating by Cell Subset

CytoNorm

CytoNorm Vignettes

CyCombine

CyCombine Vignettes

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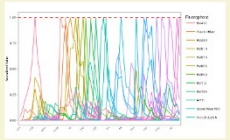
Pipeline: Clustering



1) Acquisition
(Cytek Aurora 5L)



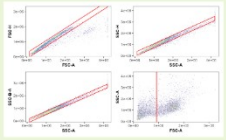
2) Unmixing Controls
(Luciernaga)



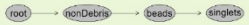
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(OLS, TRU-OLS, AutoSpectral)



4) File Checks
(PeacoQC, Flow AI, etc)



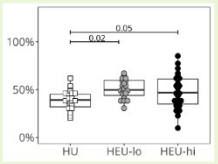
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(FlowJo or flowWorkspace)



6) Initial Gates
(openCyto or flowMagic)



7) FlowJo .wsp
(CytoML Docker)



8) Major subset Outputs

Iterating by Experiment



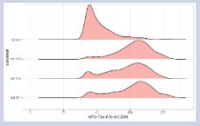
9) Subset Export



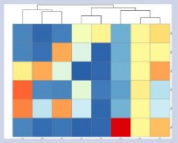
10) Return to R
(flowWorkspace)



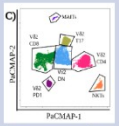
11) Coereba



12) Normalization
(CytoNorm or CyCombine)



13) Clustering
(FlowSOM)



14) Dimensionality Visualization
(UMAP or PaCMAP)



Summarized Experiment



15) Coereba

Iterating by Cell Subset



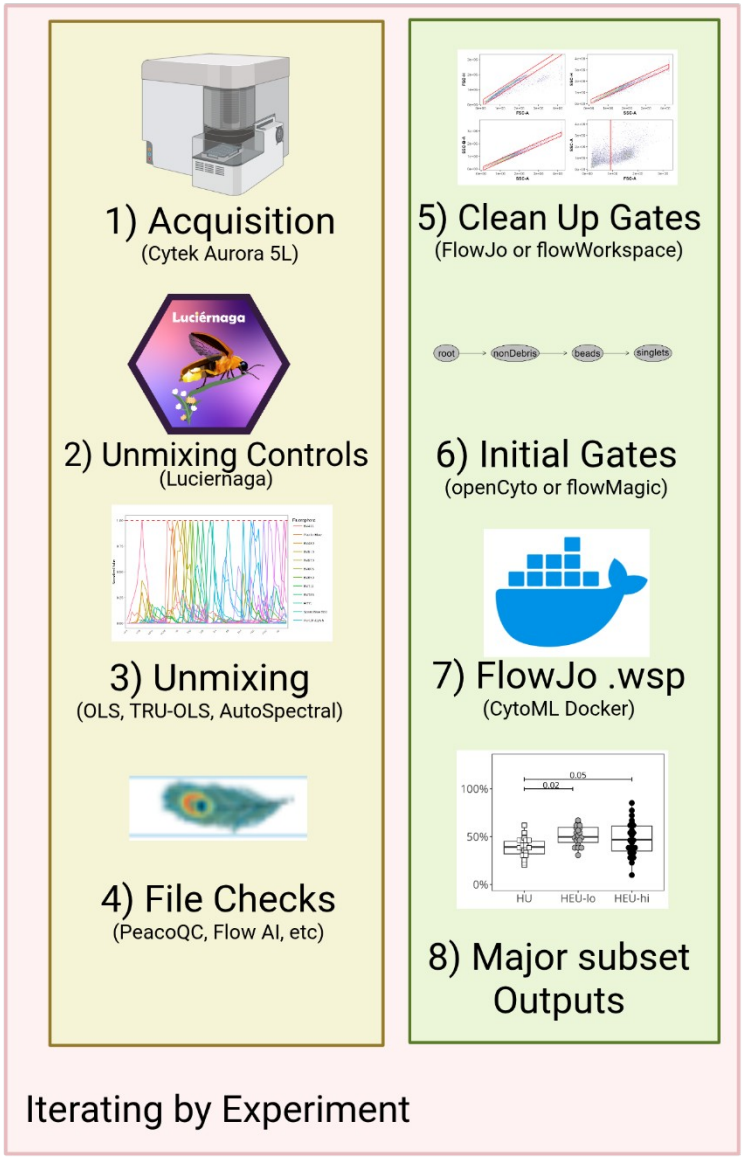
FlowSOM

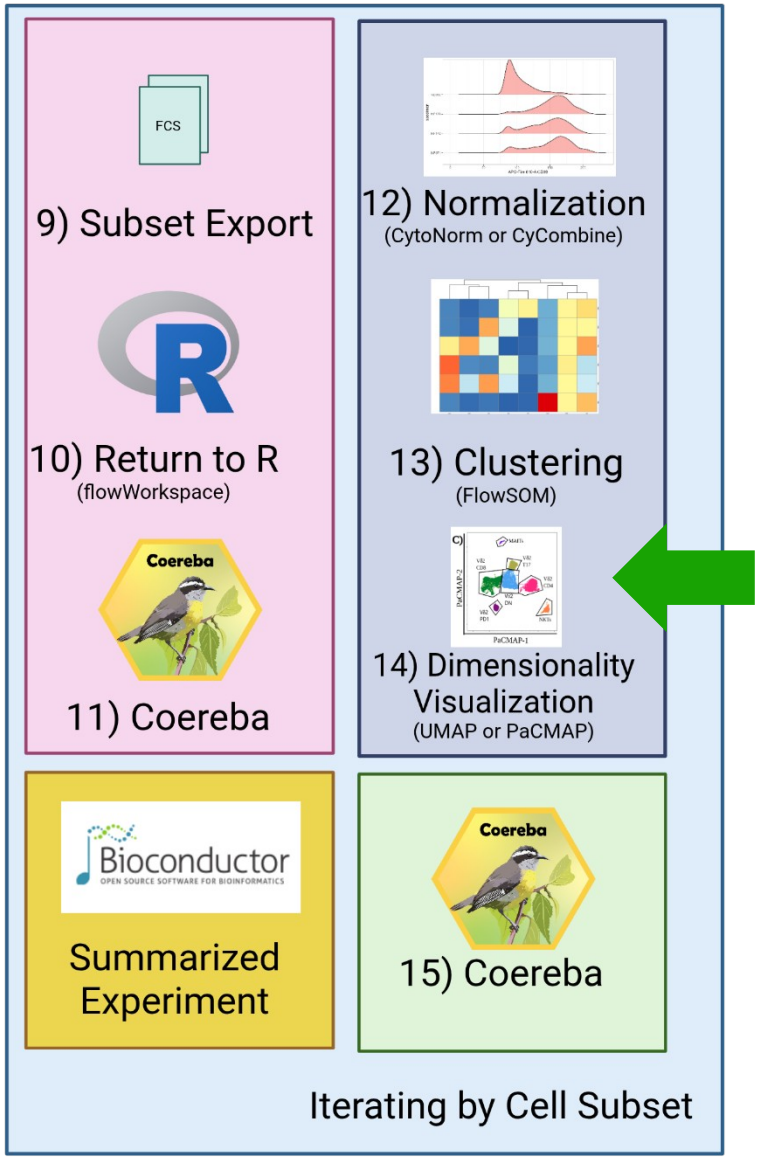


FlowSOM Vignettes

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Pipeline: Dimensionality Visualization





UMAP

UMAP Wrapper

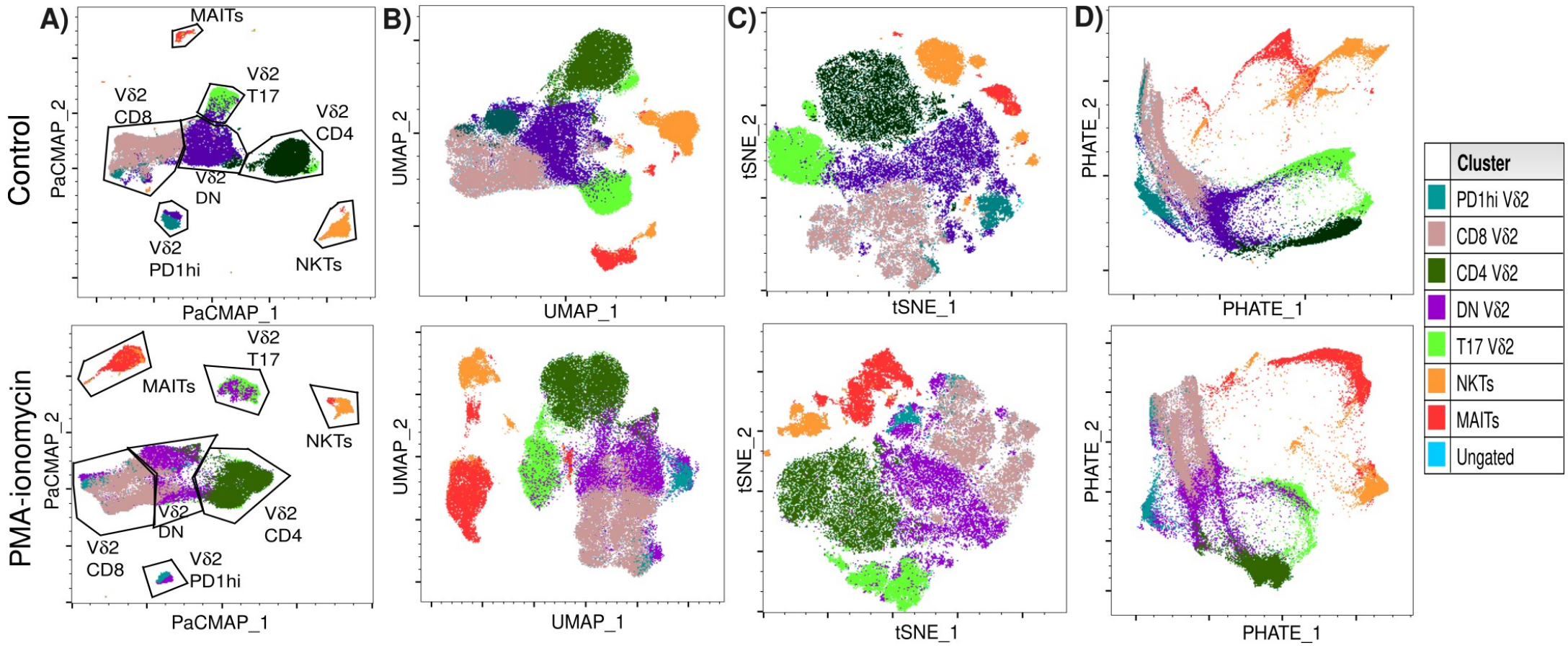
PaCMAP

PaCMAP Wrapper

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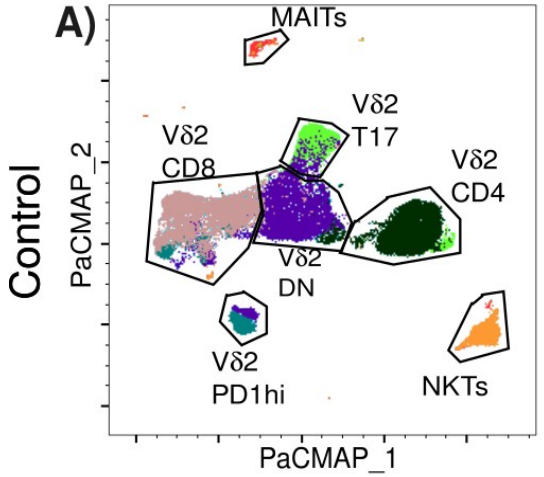
Ability to contrast unsupervised methods vs. “manual” ground-truth



University of Maryland School of Medicine

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Retrieving and re-enumerating

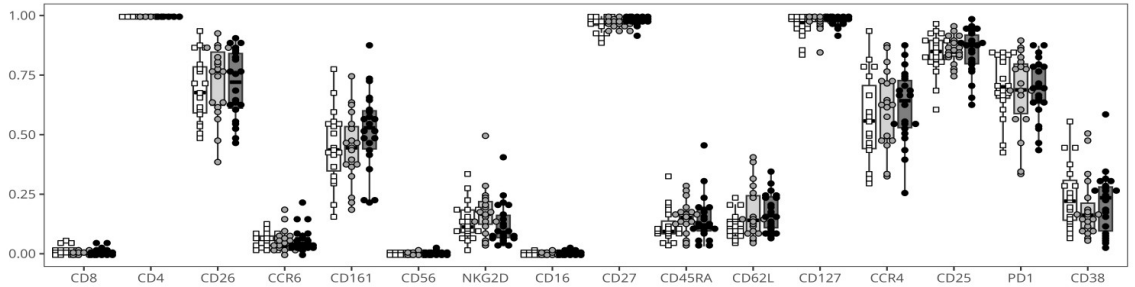


1) Gate

2) Retrieve manual-annotated gates

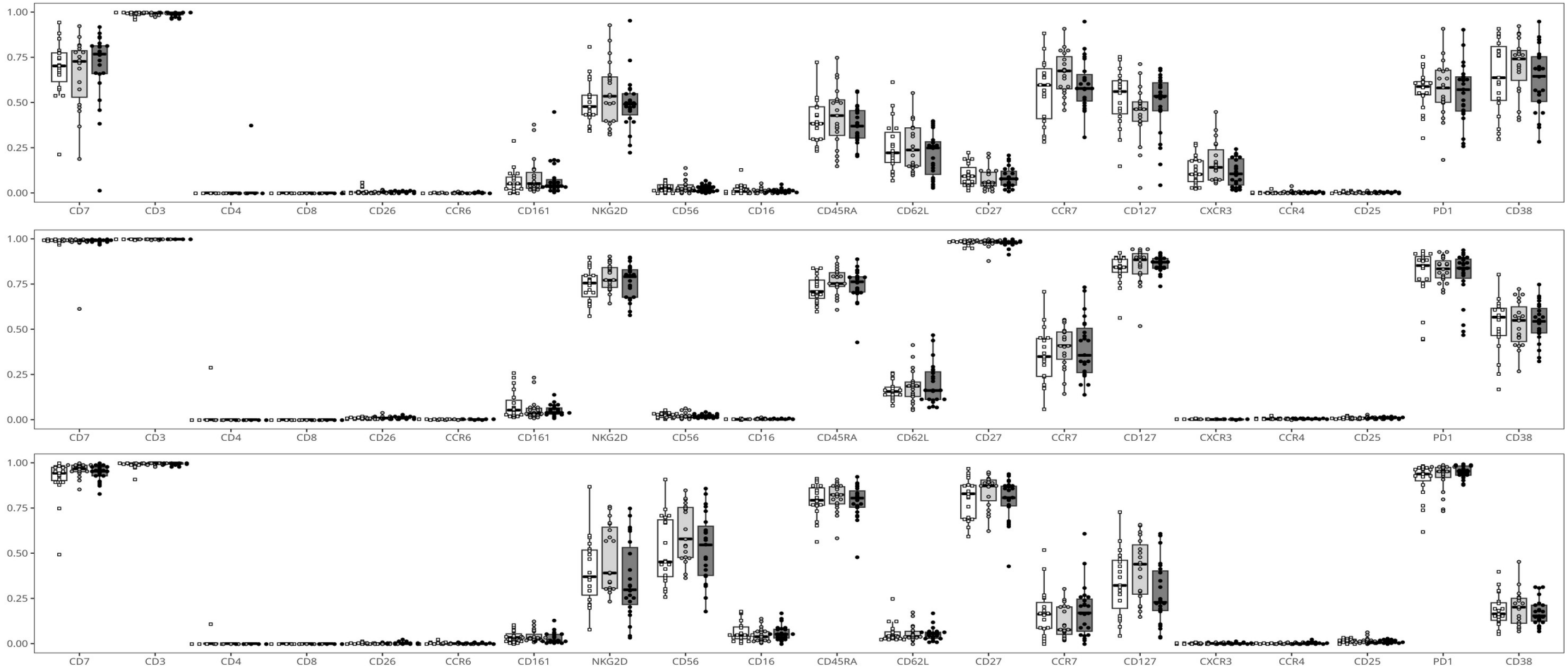
PEVio770pos	APCneg	AlexaFluor647neg	APCR700neg	APCFire750neg	APCFire810neg
PEVio770pos	APCneg	AlexaFluor647neg	APCR700neg	APCFire750neg	APCFire810neg
PEVio770pos	APCpos	AlexaFluor647neg	APCR700neg	APCFire750pos	APCFire810neg
PEVio770pos	APCneg	AlexaFluor647pos	APCR700neg	APCFire750neg	APCFire810pos

3) Re-enumerate and plot



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For each island, re-aggregate cells from each donor



DN T cells

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Cord blood T cells - Baseline

Marker	1	2	3	4	5	6	7	8
CD7	+++	+++	+++	+++	+++	+++	+++	+++
CD3	+++	+++	+++	+++	+++	+++	+++	+++
VD2	-	-	-	-	-	-	-	-
Va7.2	-	-	-	-	-	-	-	+++
Va24Ja18	-	-	-	-	-	-	-	-
CD8	-	-	-	-	-	-	-	-
CD4	+++	+++	+++	+++	+++	+++	+++	+++
CD26	-/+	++	+	++	+++	+++	+++	+++
CCR6	-	-	-	-	-	-	-	-
CD161	-	+	-	-	-	-	++	-
NKG2D	-	-	-/+	-	-	-	+++	-
CD56	-	-	-	-	-	-	+++	-
CD16	-	-	-	-	-	-	+	-
CD62L	-/+	-	-	+++	-	+++	+	+
CD27	+++	+++	+	+++	+++	+++	+++	+++
CD45RA	+++	+	+++	+++	+++	+++	+++	+++
CD127	-	+++	+++	+++	+++	+++	+++	+++
CD25	+++	+++	-	-	-	-	-	-
CCR4	-	+++	-	-	-	-	-	-
PD1	-/+	+	-	-	-	-	-	-
CD38	+++	+	+++	+++	+++	+++	+++	+++

CD4 T cells

Marker	1	2	3	4	5	6	7	8	9	10	11
CD7	+++	+++	+++	+++	+++	+++	+++	+++	+++	++	+++
CD3	+++	+++	+++	+++	+++	+++	+++	+++	+++	+++	+++
VD2	-	-	-	-	-	-	-	-	-	-	-
Va7.2	+++	-	-	-	-	-	-	-	-	-	-
Va24Ja18	-	-	-	-	-	-	-	-	-	-	-
CD8	+++	+++	+++	+++	+++	+++	+++	+++	+++	+++	+++
CD4	-	-	-	-	-	-	-	-	-	-	-
CD26	++	-	+	++	-/+	+++	+++	-	-	-	-
CCR6	-	-	-	-	-	-	+++	-	-	-	-
CD161	-	++	-	-	-	+++	+++	-	-	-/+	+++
NKG2D	+++	+++	+++	+++	+++	+++	+++	+++	-	+++	+++
CD56	-	-	-	-	-	-	-	-	-/+	-	+++
CD16	-	+++	-	-	-	-	-	-	-	-	+++
CD62L	-	-	+++	-	++	-	-	-/+	-	-	-
CD27	+++	+	+++	+++	+	+++	+++	+++	+++	+	-/+
CD45RA	+++	+++	+++	+++	+++	+++	+++	++	++	+++	+++
CD127	+++	-	+++	+++	+++	+++	+++	+++	-/+	-/+	-
CD25	-	-	-	-	-	-	+	-	-	-	-
CCR4	-	-	-	-	-	-	+	-	-	-	-
PD1	-	++	-	-	-	-	-	+	+++	+	-
CD38	+++	+++	+++	+++	+++	+++	-	-/+	+	++	+++

CD8 T cells

Marker	1	2	3	4	5
CD7	++	+++	+++	+++	+++
CD3	+++	+++	+++	+++	+++
VD2	-	-	-	-	-
Va7.2	-	-	-	-	-
Va24Ja18	-	-	-	-	-
CD8	-	-	-	-	-
CD4	-	-	-	-	-
CD26	-	-	-	+++	-
CCR6	-	-	-	+++	-
CD161	-	-	-	+++	+++
NKG2D	++	+++	+	+++	+++
CD56	-	-	++	-	+++
CD16	-	-	-	-	+++
CD62L	-/+	-/+	-	-	-
CD27	-	+++	+++	+++	-/+
CD45RA	+	+++	+++	+++	+++
CD127	++	+++	+	+++	-
CD25	-	-	-	++	-
CCR4	-	-	-	++	-
PD1	++	+++	+++	-/+	-
CD38	++	++	-/+	-	+++

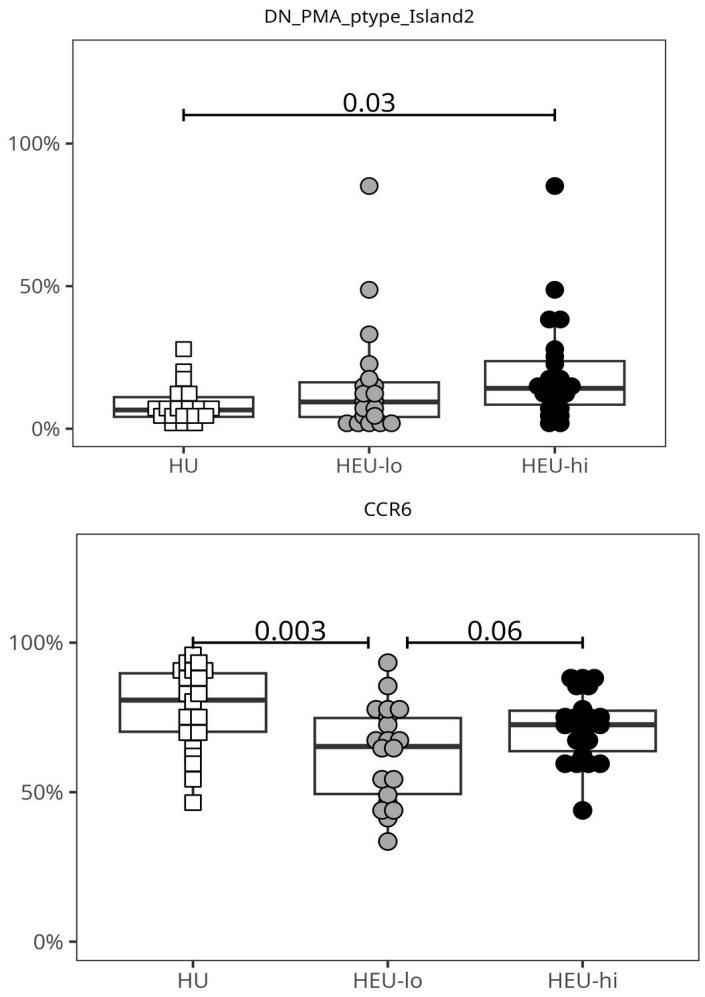
DN T cells

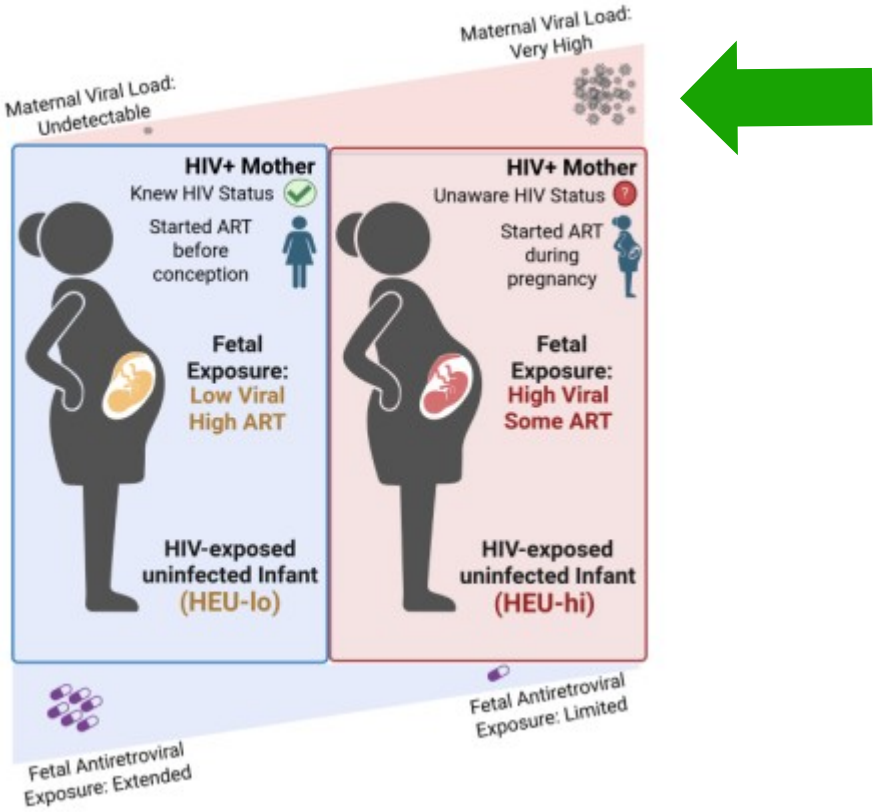
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CD4-CD8- Cord blood T cell subsets driving observed IFNɣ increases

Marker	1	2	3	4
CD7	+++	+++	+++	+++
CD3	+++	+++	+++	+++
VD2	-	-	-	-
Va7.2	-	-	-	-
Va24Ja18	-	-	-	-
CD8	-	-	-	-
CD4	-	-	-	-
CD26	+++	-	-	-
CCR6	++	-	-	-
CD161	+++	+++	-	-
NKG2D	+++	+++	+++	-
CD56	-	++	-	-/+
CD16	-	-	-	-
CD62L	-	-	-	-
CD27	+++	-	+++	++
CD45RA	+++	+++	+++	+++
CD127	+++	-	+++	-
CD25	++	-	-	-
CCR4	-	-	-	-
PD1	++	-/+	+++	+++
CD38	-	+++	++	+
CD69	+++	+++	+++	+++
TNFα	+	++	-	-
IFNγ	-	++	-	-/+
CD107a	-	-	-	-



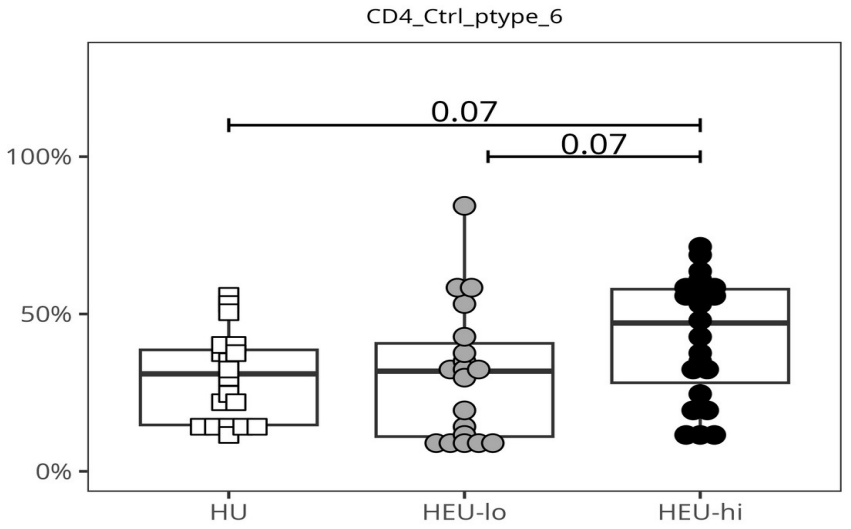


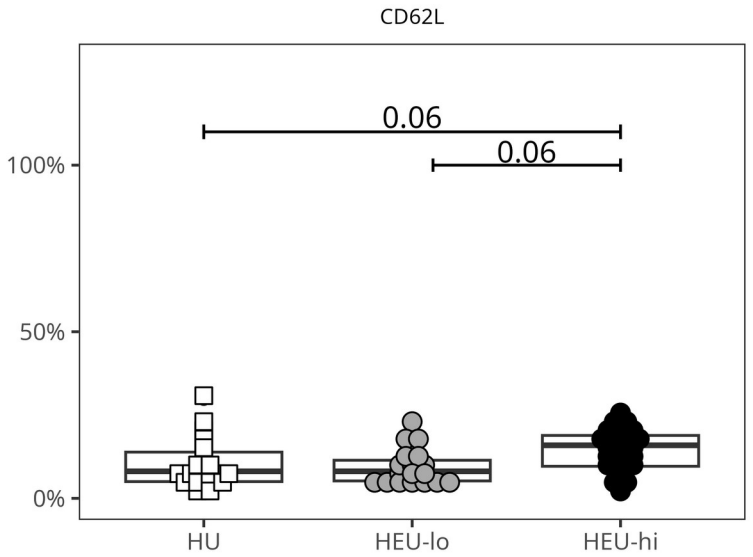
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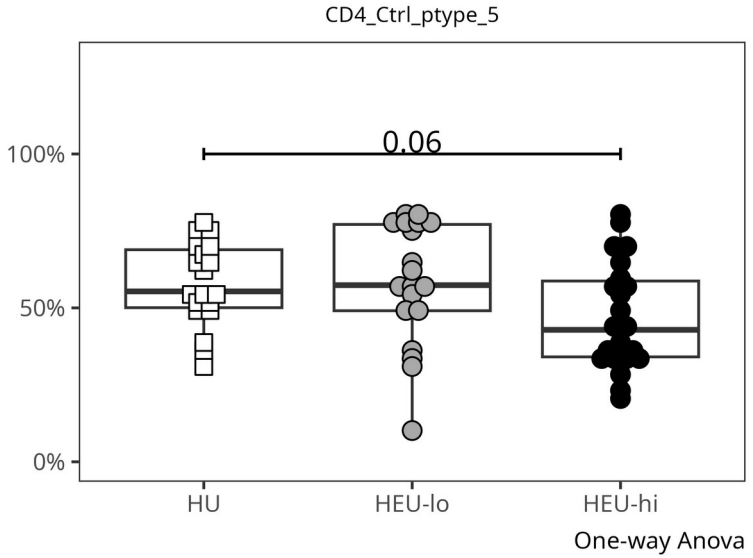
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Conventional Cord blood T cell subsets also show changes in CD62L

Marker	1	2	3	4	5	6	7	8
CD7	+++	+++	+++	+++	+++	+++	+++	+++
CD3	+++	+++	+++	+++	+++	+++	+++	+++
VD2	-	-	-	-	-	-	-	-
Va7.2	-	-	-	-	-	-	-	+++
Va24Ja18	-	-	-	-	-	-	-	-
CD8	-	-	-	-	-	-	-	-
CD4	+++	+++	+++	+++	+++	+++	+++	+++
CD26	-/+	++	+	++	+++	+++	+++	+++
CCR6	-	-	-	-	-	-	-	-
CD161	-	+	-	-	-	-	++	-
NKG2D	-	-	-/+	-	-	-	+++	-
CD56	-	-	-	-	-	-	+++	-
CD16	-	-	-	-	-	-	+	-
CD62L	-/+	-	-	+++	-	+++	+	+
CD27	+++	+++	+	+++	+++	+++	+++	+++
CD45RA	+++	+	+++	+++	+++	+++	+++	+++
CD127	-	+++	+++	+++	+++	+++	+++	+++
CD25	+++	+++	-	-	-	-	-	-
CCR4	-	+++	-	-	-	-	-	-
PD1	-/+	+	-	-	-	-	-	-
CD38	+++	+	+++	+++	+++	+++	+++	+++

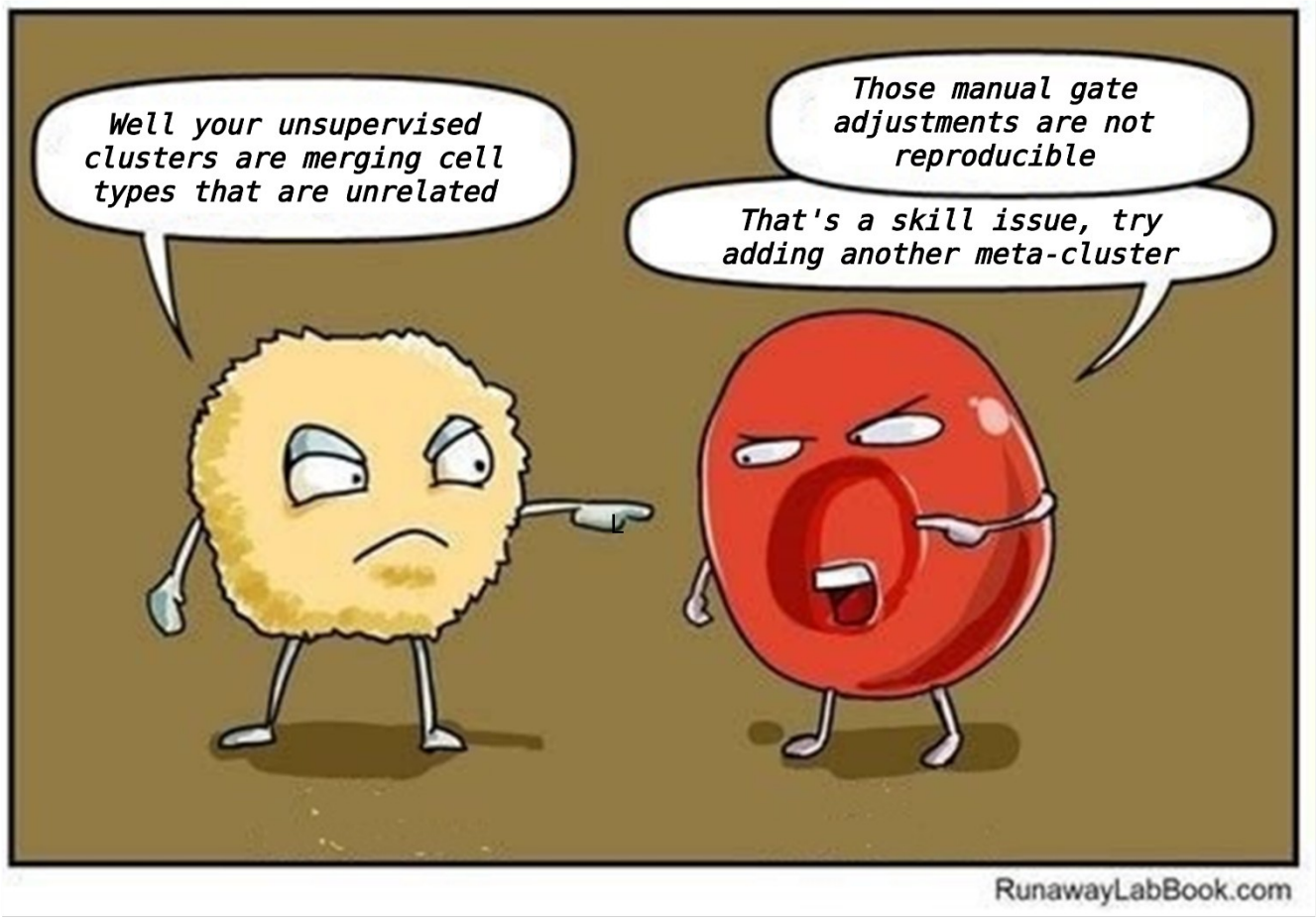






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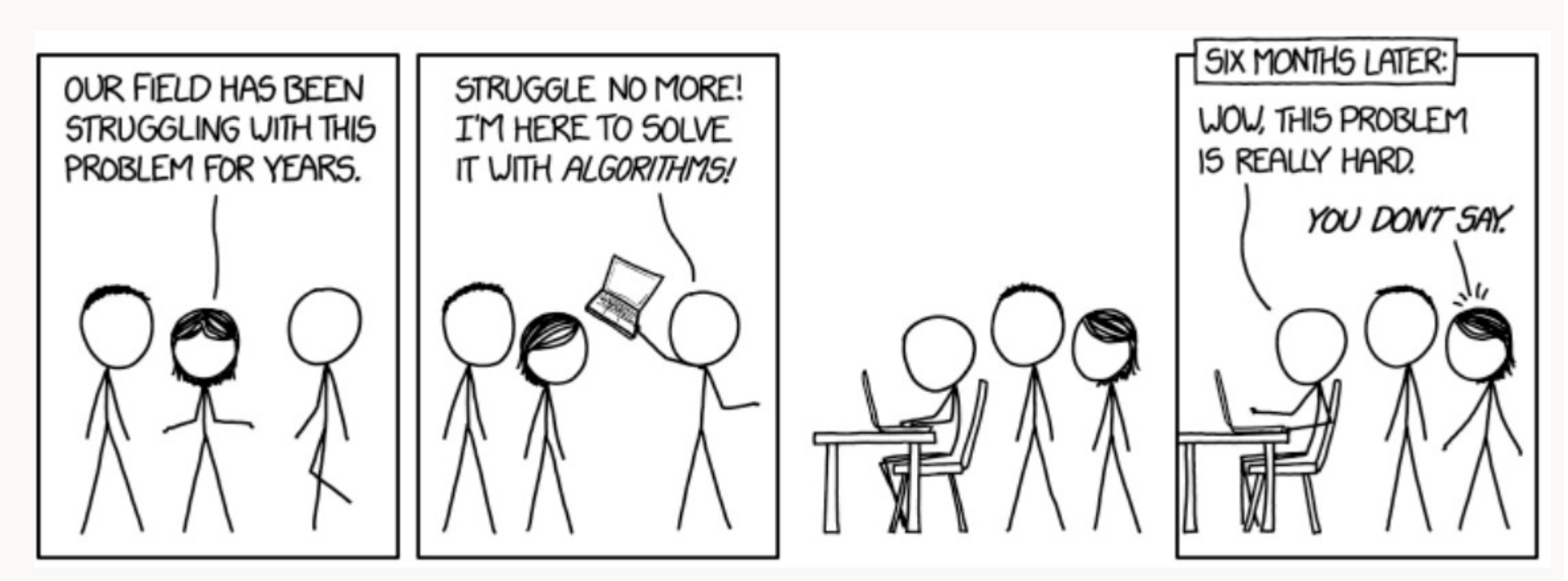
Functional and Iterative: plenty for everyone to dislike :)



Fun off-label uses thus far:

- Contrasting where “manually-gated” cells end up vs. our unsupervised methods (normalization, clustering, etc.)
- Added benefit of looking-at-the-forest: Easy to identify badly cryopreserved specimens where cell populations were lost

Are our high-dimensional analysis problems fixed?



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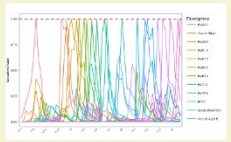
Semi-supervised Pipeline



1) Acquisition
(Cytek Aurora 5L)



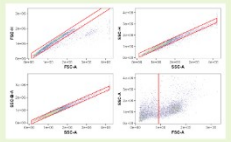
2) Unmixing Controls
(Luciernaga)



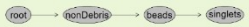
3) Unmixing
(OLS, TRU-OLS, AutoSpectral)



4) File Checks
(PeacoQC, Flow AI, etc)



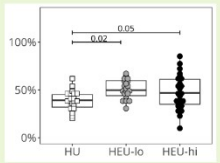
5) Clean Up Gates
(FlowJo or flowWorkspace)



6) Initial Gates
(openCyto or flowMagic)



7) FlowJo .wsp
(CytoML Docker)



8) Major subset Outputs

Iterating by Experiment



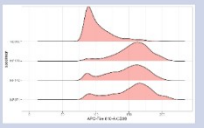
9) Subset Export



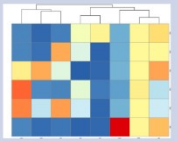
10) Return to R
(flowWorkspace)



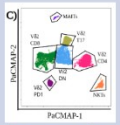
11) Coereba



12) Normalization
(CytoNorm or CyCombine)



13) Clustering
(FlowSOM)



14) Dimensionality Visualization
(UMAP or PaCMAP)



Summarized Experiment



15) Coereba

Iterating by Cell Subset

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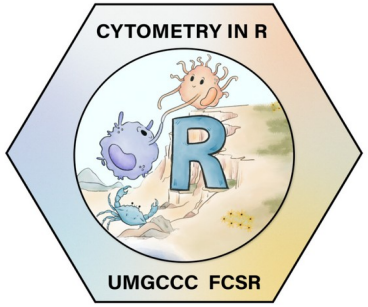
What good is any tool if the community is not able to use it?

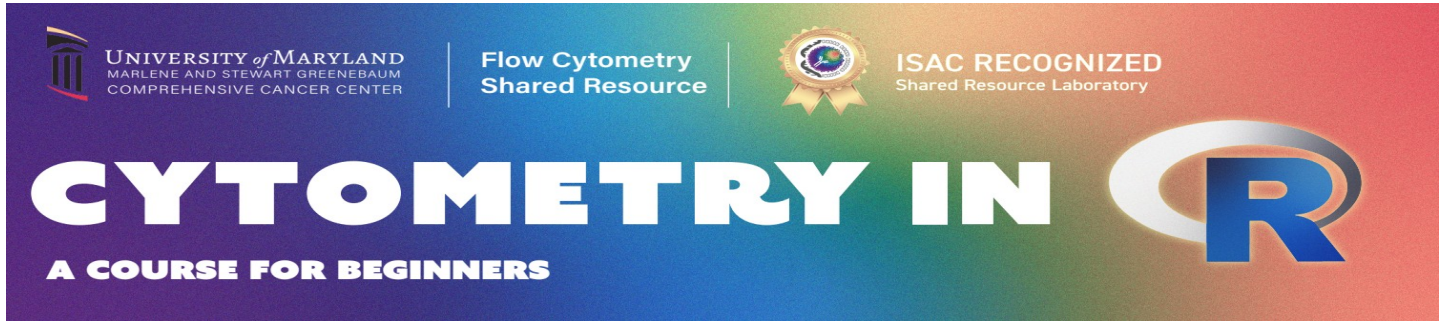
- Few resources to learn Cytometry in R to begin with
- Those that exist aimed at those with intermediate-advanced R skills, not beginners
- Know-how concentrated at a few institutions
- Most self-learning attempts end in frustration



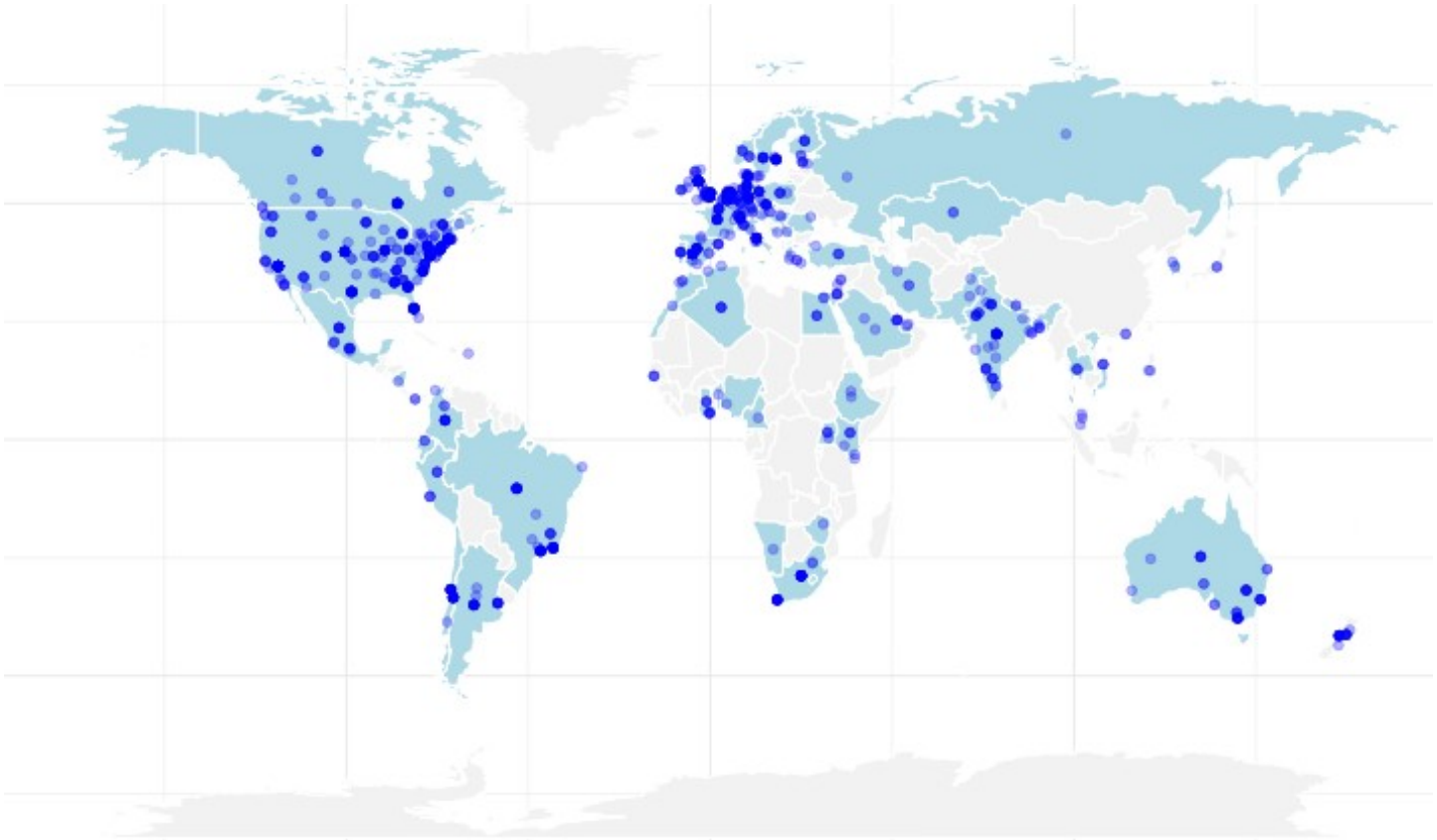
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Cytometry in R: A free weekly course for flow cytometrist with no-to-little coding experience





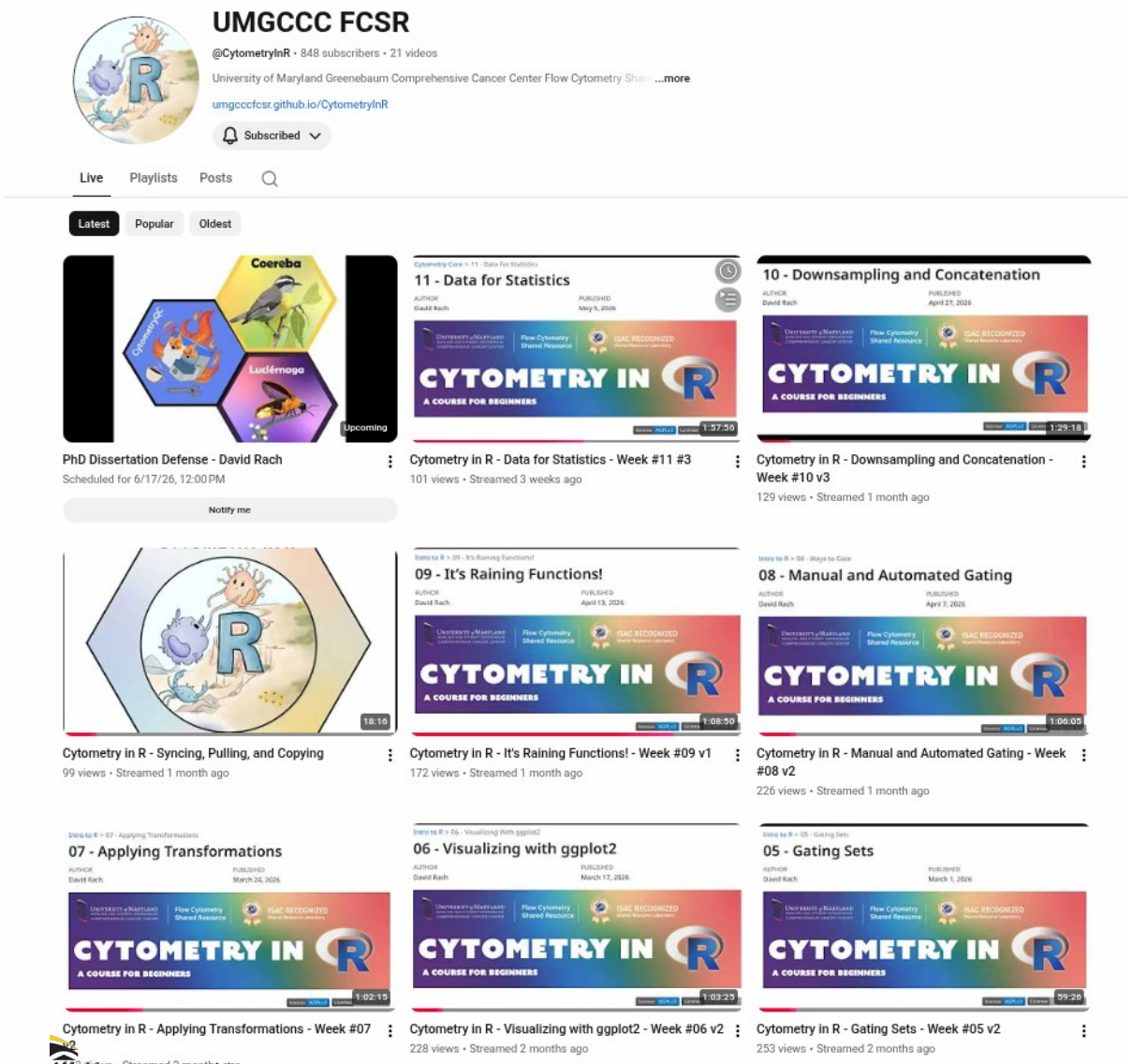
Website



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YouTube

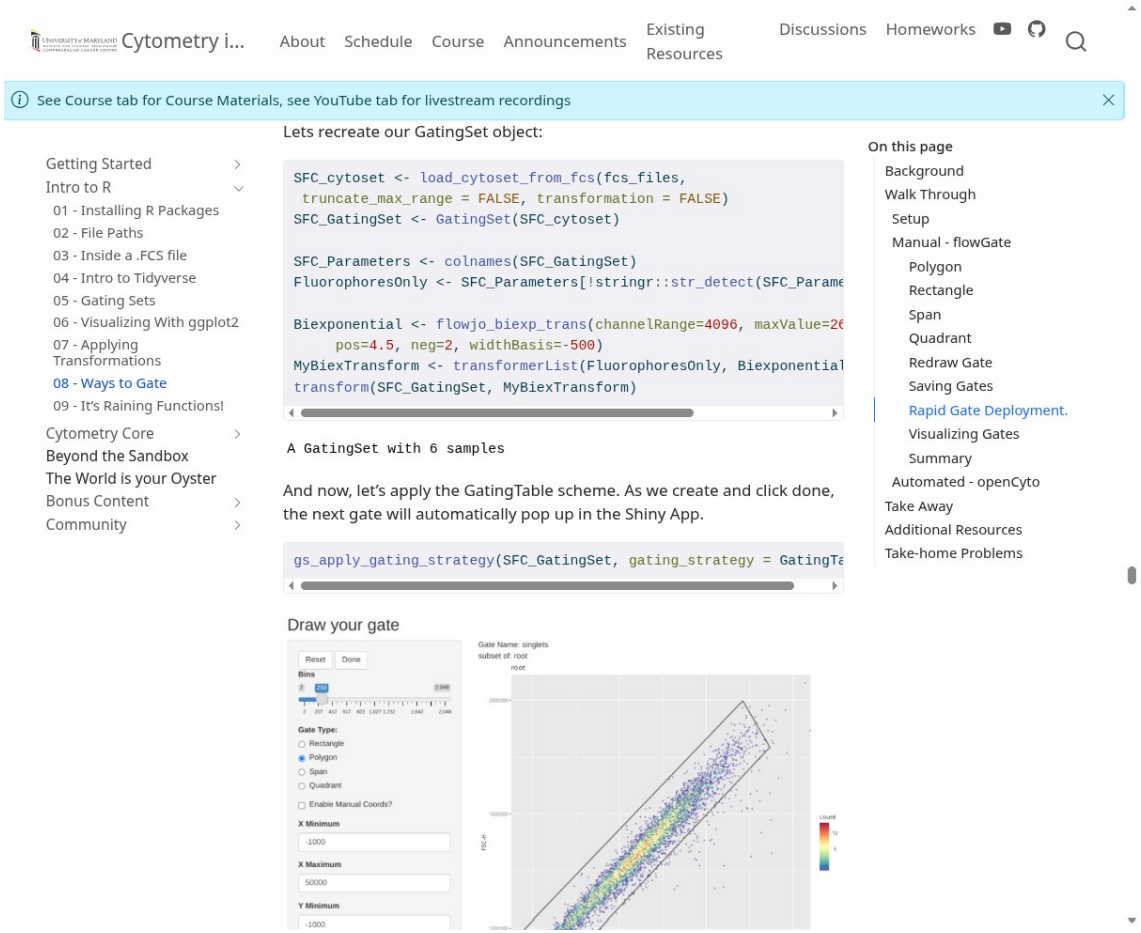


- Three live-streams a week (accommodating different timezones and schedules)
- Recordings available immediately after
- Best of the weekly livestreams added to the playlist, cutting down on editing

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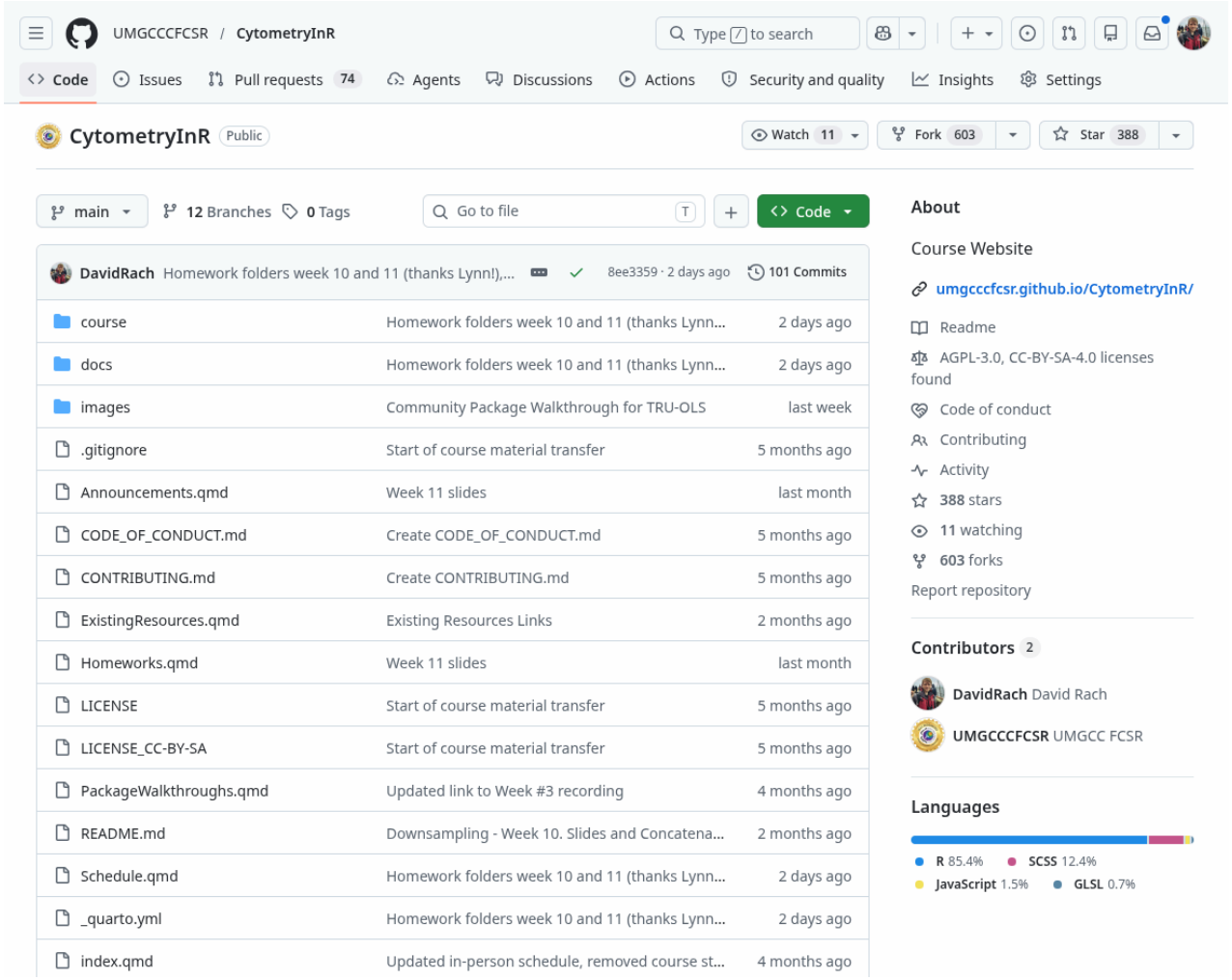
Website

- Created with Quarto, hosted via GitHub pages.
- Code, code outputs, screenshots and explanations
- Searchable
- New weekly material added as a new .qmd file



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GitHub

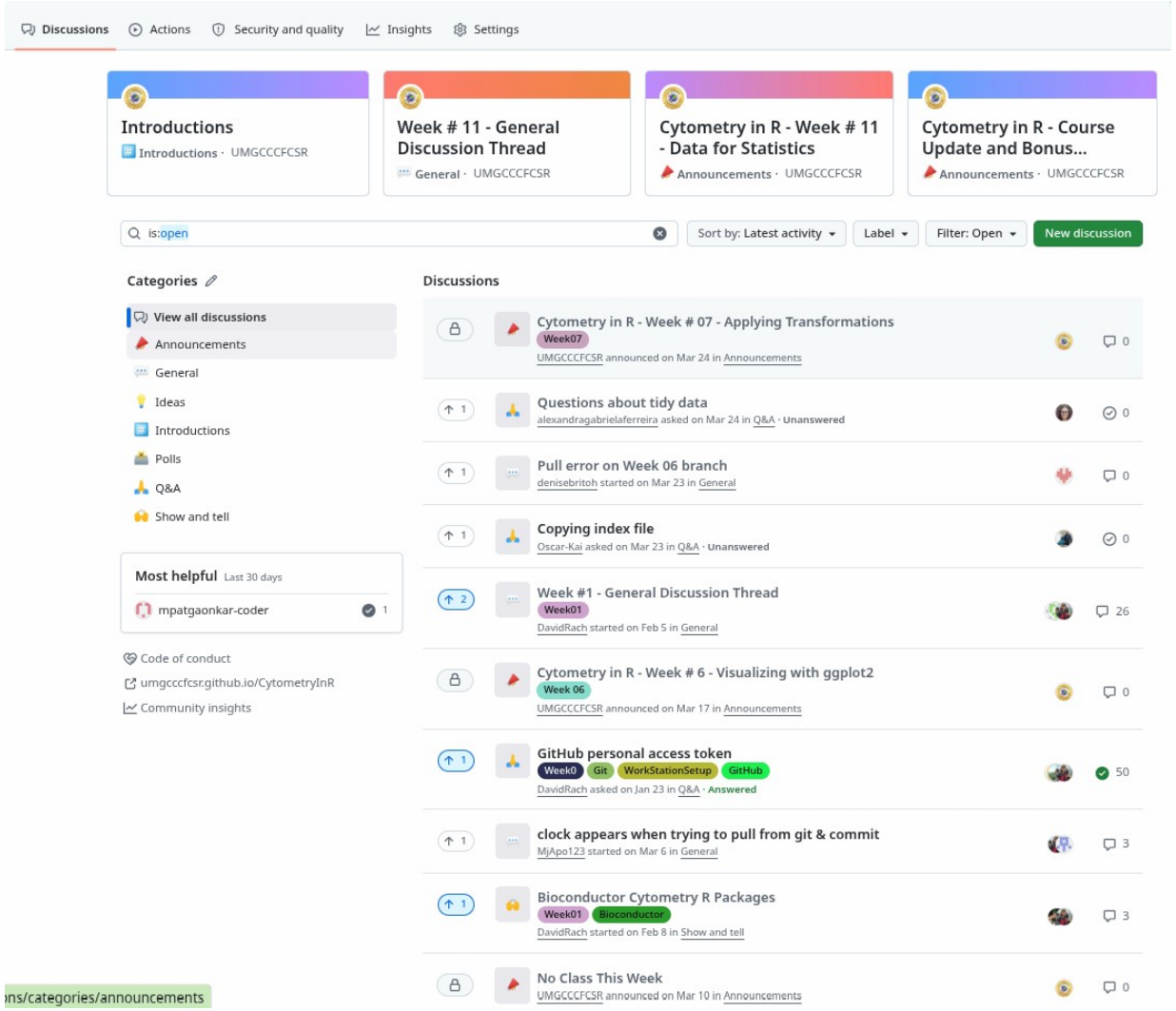


- Code and datasets available via the GitHub repository
- Course participants start course by creating GitHub account and forking the repository
- After Positron is linked to GitHub account, pull locally
- When new material posted each week, sync and pull

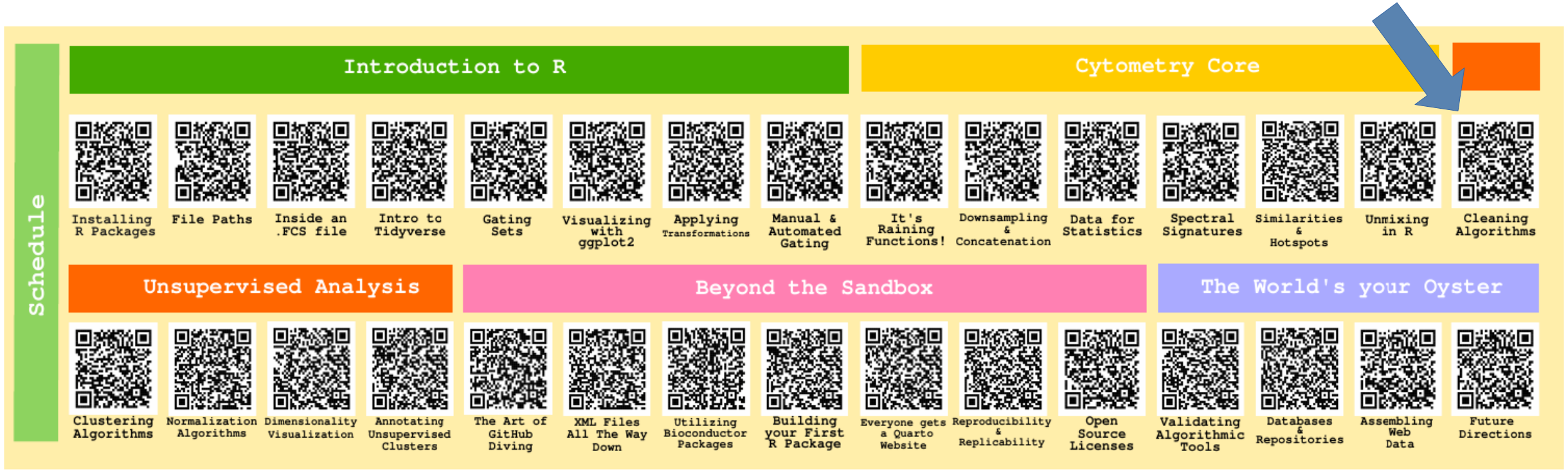
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Discussion Forum

- GitHub Discussions page has been repurposed as a community forum for questions, etc.



Half-way through this passion project



- Wrapping up Cytometry Core module
- Gathering steam to start on the Unsupervised Analysis section

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Acknowledgments

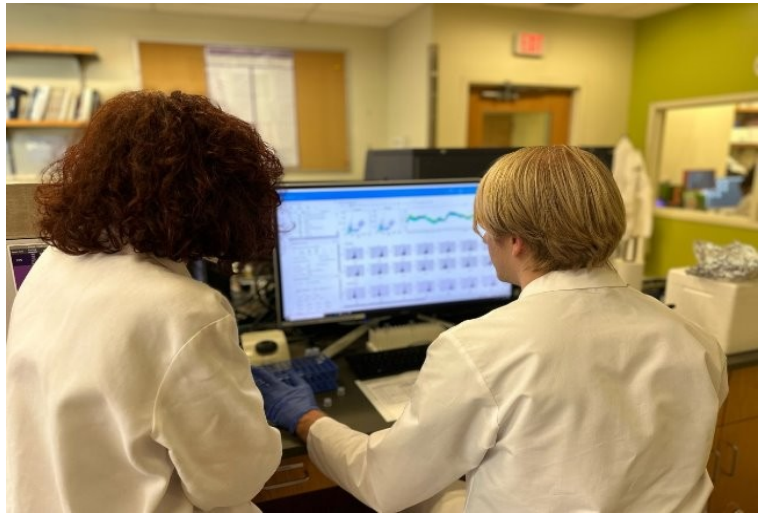
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OF HEALTH SCIENCES

All mothers and their infants, and their communities

All the open-source contributors, developers, and maintainers, that collectively made the tool sets without which all of this would not be feasible

Dr. Cristiana Cairo
Dr. Kirsten E. Lyke
Dr. Xiaoxuan Fan

Dr. Daniela Franco
Dr. Natarajan Ayithan



U01 HD092309

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Cytometry Discord

Bioconductor Zulip

https://github.com/DavidRach

@davidrach.bsky.social

www.linkedin.com/in/david-rach/

Want to collaborate? Reach out!

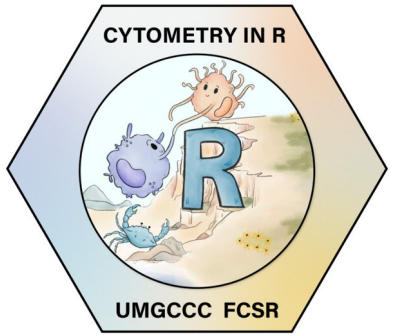
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MARLENE AND STEWART GREENEBAUM
COMPREHENSIVE CANCER CENTER

ISAC RECOGNIZED
Shared Resource Laboratory

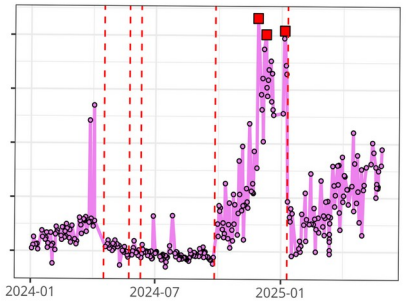
Questions?



Luciernaga



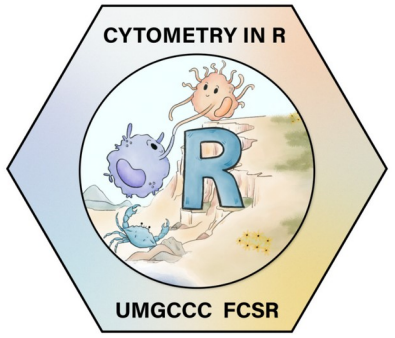
Website - CytometryInR



UMGCCC FCSR
InstrumentQC dashboard



Coereba



YouTube @**CytometryInR**



InstrumentQC How-To

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Biology is (fortunately) not just math



- **29-color** SFC panel
  → **536 million (theory)**

When actually exploring our panels dimensional space

CD4 T cells → 16241

CD8 T cells → 12466

DN T cells → 9035

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